

SEQUENCE AND SERIES



SYNOPSIS

Sequence:

A set of numbers is arranged in a definite order according to some definite rule is called a sequence. e.g. 2, 4, 6, 8,, is a sequence

- A sequence is a function whose domain is a set of natural numbers. If the range of a sequence is a subset of real numbers (or complex numbers), then it is called a real sequence (or complex sequence)

Series:

The sum of the terms of a sequence is called a series.

- If $a_1, a_2, a_3, \dots$ is a sequence, then the expression $a_1 + a_2 + a_3 + \dots$ is a series
- A series is called finite series, if it has finite number of terms. Otherwise it is called infinite series.
e.g. i) $1+3+5+\dots+21$ is a finite series.
ii) $2+4+6+8+\dots$ is an infinite series.
- Sequences following specific patterns are called progressions.

Arithmetic progression (A.P):-

- A sequence is called an arithmetic progression, if the difference between any two consecutive terms is the same.
- A.P is of the form $a, a+d, a+2d, a+3d, \dots$
where a is 1st term and d is common difference

General term of an A.P:

- Let 'a' be the first term and 'd' be common difference of an A.P, then its general term (or) n^{th} term is $T_n = a + (n-1)d$
- If 'l' be the last term and 'd' be common difference of an A.P, then m^{th} term from the end $T'_m = l - (m-1)d$
- m^{th} term from the end = $(n-m+1)^{\text{th}}$ term from the beginning.

W.E-1: Find the first negative term of the sequence

$$20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots$$

Sol: The given sequence is an A.P in which first term $a=20$ and common difference $d=-3/4$.

Let the n^{th} term of the given A.P. be the first negative term. Then, $a_n < 0$

$$\Rightarrow a + (n-1)d < 0 \Rightarrow 20 + (n-1)(-3/4) < 0$$

$$\Rightarrow \frac{83}{4} - \frac{3n}{4} < 0 \Rightarrow 83 - 3n < 0 \Rightarrow 3n > 83$$

$$\Rightarrow n > 27\frac{2}{3} \Rightarrow n = 28$$

thus, 28th term of the given sequence is the first negative term.

W.E-2: If 100 times the 100th term of an A.P with non-zero common difference equals the 50 times of 50th term, then find 150th term of this A.P. (AIEEE 2012)

Sol: $100T_{100} = 50T_{50}$; $100(a+99d)=50(a+49d)$
 $2a+198d=a+49d$; $a+149d=0$
 $T_{150} = a + 149d = 0$

Sum to n terms of an A.P:

$$S_n = \frac{n}{2}[a+l] = \frac{n}{2}[2a+(n-1)d]$$

where a = first term, l = last term

d = common difference

- If the sum of n terms of a sequence S_n is given, then its n^{th} term T_n can be determined by $T_n = S_n - S_{n-1}$

W.E-3: How many terms are to be added to make the sum 52 in the series $(-8)+(-6)+(-4)+\dots$?

Sol: $S_n = 52 \Rightarrow \frac{n}{2}[2(-8)+(n-1)2] = 52$

$$\Rightarrow n(2n-18) = 104$$

$$\Rightarrow n(n-9) = 52 \Rightarrow n = 13$$

W.E-4: Let $a_1, a_2, \dots, a_n$ be the terms of an A.P.

If $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2}{q^2}$, $p \neq q$ then find $\frac{a_6}{a_{12}}$.

Sol:

$$\begin{aligned} \frac{\frac{p}{2}[2a_1 + (p-1)d]}{\frac{q}{2}[2a_1 + (q-1)d]} &= \frac{p^2}{q^2} \Rightarrow \frac{[2a_1 + (p-1)d]}{[2a_1 + (q-1)d]} = \frac{p}{q} \\ \Rightarrow \frac{a_1 + \left(\frac{p-1}{2}\right)d}{a_1 + \left(\frac{q-1}{2}\right)d} &= \frac{p}{q} \text{ For } \frac{a_6}{a_{12}}, p=11, q=41 \\ \Rightarrow \frac{a_6}{a_{12}} &= \frac{11}{41} \end{aligned}$$

Properties of A.P.:-

- a, b, c are in AP $\Leftrightarrow 2b = a + c$
- In a finite A.P, the sum of the terms equidistant from the beginning and the end is always same and is equal to the sum of the first and last term
i.e., $a_2 + a_{n-1} = a_3 + a_{n-2} = a_4 + a_{n-3} = \dots = a_1 + a_n$
- $(a_1 + a_2 + a_3 + \dots + a_n)$

$$= \begin{cases} n \times (\text{middle term}), & \text{if } n \text{ is odd} \\ \frac{n}{2} \times (\text{sum of two middle terms}), & \text{if } n \text{ is even} \end{cases}$$
- If $a_1, a_2, a_3, \dots, a_n$ are in A.P then
 - $a_n, a_{n-1}, \dots, a_3, a_2, a_1$ are in A.P
 - $a_1 \pm \lambda, a_2 \pm \lambda, a_3 \pm \lambda; \dots, a_n \pm \lambda$ are in A.P (where $\lambda \in R$)
 - $\lambda a_1, \lambda a_2, \lambda a_3; \dots, \lambda a_n$ are in A.P (where $\lambda \in R - \{0\}$)
- p^{th} term of an A.P. is 'q' and q^{th} term is 'p', then $T_{p+q} = 0$
- If m^{th} term of an A.P. is 'n' and n^{th} term is 'm' then p^{th} term is 'm+n-p'
- If $S_p = q$ and $S_q = p$ for an A.P., then $S_{p+q} = -(p+q)$

W.E-5: If $1, \frac{1}{2} \log_3 (3^{1-x} + 2), \log_3 (4 \cdot 3^x - 1)$ are in A.P, then find x .

Sol: $1, \frac{1}{2} \log_3 (3^{1-x} + 2), \log_3 (4 \cdot 3^x - 1)$ are in A.P

$$\begin{aligned} \Rightarrow \log_3 (3^{1-x} + 2) &= 1 + \log_3 (4 \cdot 3^x - 1) \\ \Rightarrow \log_3 (3^{1-x} + 2) &= \log_3 3 + \log_3 (4 \cdot 3^x - 1) \\ \Rightarrow \log_3 (3^{1-x} + 2) &= \log_3 3(4 \cdot 3^x - 1) \\ \Rightarrow (3^{1-x} + 2) &= 3(4 \cdot 3^x - 1) \\ \Rightarrow 3 \cdot 3^{-x} + 2 &= 12 \cdot 3^x - 3 \\ \Rightarrow \frac{3}{t} + 2 &= 12t - 3, \text{ (where } t = 3^x) \\ \Rightarrow 3 + 2t &= 12t^2 - 3t \Rightarrow 12t^2 - 5t - 3 = 0 \\ \Rightarrow (4t - 3)(3t + 1) &= 0 \\ \Rightarrow t = \frac{3}{4}, \frac{-1}{3} \Rightarrow 3^x &= \frac{3}{4} (\because 3^x > 0) \\ \Rightarrow x &= \log_3 \left(\frac{3}{4} \right) = 1 - \log_3 4 \end{aligned}$$

Selection of terms in an A.P:

Number of terms	Terms	c.d
3	$a-d, a, a+d$	d
4	$a-3d, a-d, a+d, a+3d$	$2d$
5	$a-2d, a-d, a, a+d, a+2d$	d
6	$a-5d, a-3d, a-d, a+d, a+3d, a+5d$	$2d$

W.E-6: If the sum of four numbers in A.P is 24 and the sum of their squares is 164 then find those numbers.

Sol: $(a-3d) + (a-d) + (a+d) + (a+3d) = 24$

$$\Rightarrow 4a = 24 \Rightarrow a = 6$$

$$(a-3d)^2 + (a-d)^2 + (a+d)^2 + (a+3d)^2 = 164$$

$$\Rightarrow 2(a^2 + 9d^2) + 2(a^2 + d^2) = 164$$

$$\Rightarrow a^2 + 5d^2 = 41 \Rightarrow 36 + 5d^2 = 41 \Rightarrow d = \pm 1$$

required numbers are 3, 5, 7, 9

Some Facts about A.P:-

- If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are two A.P's then
 - a) $a_1 \pm b_1, a_2 \pm b_2, \dots$ are in AP
 - b) $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ and $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots$ are not in A.P
 - c) If the terms of an A.P. are chosen at regular intervals, then they form an A.P.
- If a constant 'k' is added to each term of A.P., with common difference 'd', then the resulting sequence also will be in A.P., with common difference (d+k).
- If every term is multiplied by a constant 'k', then the resulting sequence will also be in A.P., with the first term 'ka' and common difference 'kd'.
- If n^{th} term of the sequence $T_n = An + B$ (i.e) [Linear expression in n] then the sequence is A.P with first term is 'A+B' and common difference A (coefficient of n)
- If sum of n terms of a sequence is $S_n = An^2 + Bn + C$ (i.e. Quadratic expression in n) then the sequence is A.P with first term is $3A+B$ and common difference is $2A$. Also in this sequence n^{th} term $T_n = 2An + (A+B)$
- If the ratio of the sums of n terms of two A.P.'s is given then the ratio of their n^{th} terms may be obtained by replacing n with $(2n-1)$ in the given ratio.
- If the ratio of n^{th} terms of two A.P.'s is given, then the ratio of the sums of their n terms may be obtained by replacing n with $\frac{n+1}{2}$ in the given ratio
- Sum of the interior angles of a polygon of 'n' sides is $(n-2)180^\circ$
- The n^{th} common term of two Arithmetic Series is (L.C.M of common difference of 1st series and 2nd series) $(n-1) + 1^{\text{st}}$ common term of both series

W.E-7: Find the n^{th} term of the sequence 5, 15, 29, 47, 69, 95, ...

Sol: The given sequence is not an A.P. but the successive differences between the various terms i.e. (15-5), (29-15), (47-29), (69-47), (95-69), ... i.e. 10, 14, 18, 22, 26, ... are in A.P

Let n^{th} term of the given sequence be $t_n = an^2 + bn + c \rightarrow (1)$ Putting $n=1, 2, 3$ in (1), we get $t_1 = a + b + c \Rightarrow a + b + c = 5 \rightarrow (2)$
 $t_2 = 4a + 2b + c \Rightarrow 4a + 2b + c = 15 \rightarrow (3)$
 $t_3 = 9a + 3b + c \Rightarrow 9a + 3b + c = 29 \rightarrow (4)$
 Solving (2), (3), (4), we get $a=2, b=4, c=-1$.
 $\therefore$ the n^{th} term of the given sequence is $t_n = 2n^2 + 4n - 1$

W.E-8: The sum of the first n terms of two A.P's are in the ratio $(2n+3):(3n-1)$. Find the ratio of 5th terms of these A.P's.

Sol: Given that $\frac{S_n}{S'_n} = \frac{2n+3}{3n-1}$

The ratio of n^{th} terms $\frac{t_n}{t'_n} = \frac{2(2n-1)+3}{3(2n-1)-1} = \frac{4n+1}{6n-4}$
 $t_5 : t'_5 = 21 : 26$

W.E-9: The interior angles of a polygon are in A.P. the smallest angle is 120° and the common difference is 5° . Find the number of sides of the polygon.

Sol: Given $a=120^\circ, d=5^\circ$

Sum of the interior angles of a polygon of n sides is $(n-2)180^\circ$

$$\therefore \frac{n}{2} [2(120) + (n-1)5] = (n-2)180$$

$$\Rightarrow n[5n + 235] = (n-2)360$$

$$\Rightarrow 5n(n+47) = (n-2)360$$

$$\Rightarrow n^2 + 47n = (n-2)72$$

$$\Rightarrow n^2 - 25n + 144 = 0 \Rightarrow (n-9)(n-16) = 0$$

$$\Rightarrow n = 9 \text{ or } 16$$

(Since neglecting $n=16$, Since that case largest angle is $[120 + (15)5] = 195$, which is not possible no longer angle of a polygon is more than 180)
 $\therefore n=9$

W.E-10: Find 12th common term of two Arithmetic Series $7+10+13+\dots$ and $4+11+18+\dots$.

Sol: The n^{th} common term of between two series = (L.C.M of common difference of 1st series and 2nd series) $(n-1) + 1^{\text{st}}$ common term of both series.
 $= (\text{L.C.M of } 3, 7) (12-1) + 25 = 21(11) + 25 = 256$

W.E-11: Find the number of common terms to the two sequences $17, 21, 25, \dots, 417$ and $16, 21, 26, \dots, 466$.

Sol: series $17, 21, 25, \dots, 417$ has common difference 4
 series $16, 21, 26, \dots, 466$ has common difference 5
 LCM of 4 and 5 is 20, the first common term is 21. Hence, the series is $21, 41, 61, \dots, 401$; which has 20 terms.

Arithmetic mean (A.M):

The Arithmetic mean A of any two numbers a and

b is given by $\frac{a+b}{2}$, where a, A, b are in AP

➤ If $a_1, a_2, a_3, \dots, a_n$ are n numbers then Arithmetic mean A of these numbers is given by

$$A = \frac{1}{n} [a_1 + a_2 + \dots + a_n]$$

➤ The n numbers $A_1, A_2, A_3, \dots, A_n$ are said to be Arithmetic means between a and b if $a, A_1, A_2, A_3, \dots, A_n, b$ are in AP

Here a = First term

$$b = (n+2)^{\text{th}} \text{ term} = a + (n+1)d$$

then, $d = \frac{b-a}{n+1}$

$$A_1 = a + \frac{b-a}{n+1}$$

$$A_2 = a + \frac{2(b-a)}{n+1}, \dots$$

$$A_n = a + \frac{n(b-a)}{n+1}$$

$$A_1 + A_2 + A_3 + \dots + A_n = n \left[\frac{a+b}{2} \right]$$

W.E-12 : If n arithmetic means are inserted between 2 and 38, then the sum of the resulting series is obtained as 200, then find the value of n .

Sol: We have

$$\frac{n+2}{2} (2+38) = 200 \Rightarrow n+2 = 10 \Rightarrow n = 8$$

Arithmetic mean of the m^{th} power : Let $a_1, a_2, \dots, a_n$ be n positive real number (not all equal)

& let m be real number then $\frac{a_1^m + a_2^m + \dots + a_n^m}{n}$

$$> \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m \quad \forall m \in \mathbb{R} - [0, 1]$$

$$< \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m \quad \forall m \in (0, 1)$$

$$= \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^m \quad \forall m \in \{0, 1\}$$

W.E-13: Prove that $\sqrt{1} + \sqrt{2} + \dots + \sqrt{n} < n\sqrt{\frac{n+1}{2}}$

Sol: $\frac{\sqrt{1} + \sqrt{2} + \dots + \sqrt{n}}{n} < \left(\frac{1+2+3+\dots+n}{n} \right)^{\frac{1}{2}}$

$$< \left(\frac{n \left(\frac{n+1}{2} \right)}{n} \right)^{\frac{1}{2}} < \left(\frac{(n+1)}{2} \right)^{\frac{1}{2}}$$

$$\therefore \sqrt{1} + \sqrt{2} + \dots + \sqrt{n} < n\sqrt{\frac{n+1}{2}}$$

Geometric Progression (G.P):- A Sequence is called a Geometric progression, if the ratio of any two consecutive terms is the same

➤ G.P is of the form $a, ar, ar^2, ar^3, \dots$, Where a is the first term and r is the common ratio

General term of G.P:- If 'a' be the first term and 'r' be the common ratio, then general term (or)

$$n^{\text{th}} \text{ term of G.P is } T_n = ar^{n-1}$$

➤ The n^{th} term from the end of a finite G.P consisting of m terms $= ar^{m-n}$

➤ The n^{th} term from the end of a finite G.P with last

term l and common ratio r is $l \left(\frac{l}{r} \right)^{n-1}$

Sum to n terms of a G.P:

➤ a) sum of n terms

$$S_n = a + ar + ar^2 + \dots + ar^{n-1} = a \left(\frac{1-r^n}{1-r} \right), \text{ if } r < 1$$

$$= a \left(\frac{r^n - 1}{r - 1} \right), \text{ if } r > 1 = na, \text{ if } r = 1$$

b) If l be the last term of the G.P., then $l = ar^{n-1}$,

$$S_n = \frac{a-lr}{1-r}, \text{ if } r < 1 = \frac{lr-a}{r-1}, \text{ if } r > 1$$

➤ If the number of terms are infinite, then the sum of

$$\text{G.P. is } S_\infty = a + ar + ar^2 + \dots = \frac{a}{1-r} \text{ if } |r| < 1$$

W.E-14: If the third term of G.P is 4, then find the product of first 5 terms.

Sol: Given $t_3 = ar^2 = 4$

Product of first 5 terms =

$$(a)(ar)(ar^2)(ar^3)(ar^4) = a^5 r^{10} = (ar^2)^5 = 4^5 = 1024$$

W.E-15: If

$(10)^9 + 2(11)^1(10)^8 + 3(11)^2(10)^7 + \dots + 10(11)^9 = k(10)^9$,
then find k . [JEE MAIN 2014]

Sol:

$$k(10)^9 = 10^9 + 2(11)^1(10)^8 + 3(11)^2(10)^7 + \dots + 10(11)^9$$

$$\Rightarrow k = 1 + 2\left(\frac{11}{10}\right) + 3\left(\frac{11}{10}\right)^2 + \dots + 10\left(\frac{11}{10}\right)^9 \rightarrow (1)$$

$$\frac{11k}{10} = \frac{11}{10} + 2\left(\frac{11}{10}\right)^2 + 3\left(\frac{11}{10}\right)^3 + \dots + 10\left(\frac{11}{10}\right)^{10} \rightarrow (2)$$

$$(1) - (2) \Rightarrow -\frac{k}{10} = 1 + \frac{11}{10} + \left(\frac{11}{10}\right)^2 + \dots + \left(\frac{11}{10}\right)^9 - 10\left(\frac{11}{10}\right)^9$$

$$= \frac{\left(\frac{11}{10}\right)^{10} - 1}{\frac{11}{10} - 1} - 10\left(\frac{11}{10}\right)^9 = 10\left(\frac{11}{10}\right)^9 - 10 - 10\left(\frac{11}{10}\right)^9$$

$$\therefore \frac{-k}{10} = -10 \Rightarrow k = 100$$

W.E-16: Three Positive numbers from an increasing G.P. If the middle term in this GP is doubled, the new number are in A.P. Then find the common ratio of the G.P. [JEE-2014]

Sol: Let a, ar, ar^2 be in G.P and $r > 1$.

Given $a, 2ar, ar^2$ are in A.P.

$$\Rightarrow 2(2ar) = a + ar^2 \Rightarrow r^2 - 4r + 1 = 0$$

$$r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3} \therefore r > 1 \Rightarrow r = 2 + \sqrt{3}$$

Selection of terms in G.P:

No. of	Terms	Common terms ratio
3	$\frac{a}{r}, a, ar$	r
4	$\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$	r^2
5	$\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$	r
6	$\frac{a}{r^5}, \frac{a}{r^3}, \frac{a}{r}, ar, ar^3, ar^5$	r^2

W.E-17: Three numbers are in G.P. Whose sum is 70, if the extremes be each multiplied by 4 and the mean by 5, they will be in A.P. then find the sum of numbers.

Sol: Let the numbers be a, ar, ar^2 and sum=70

$$\Rightarrow a(1+r+r^2) = 70 \rightarrow (1)$$

it is given that $4a, 5ar, 4ar^2$ are in A.P

$$\Rightarrow 2(5ar) = 4a + 4ar^2 \Rightarrow 5r = 2 + 2r^2$$

$$\Rightarrow 2r^2 - 5r + 2 = 0 \Rightarrow (2r-1)(r-2) = 0$$

$$\Rightarrow r = 2, \frac{1}{2} \text{ put } r=2 \text{ in } (1), \text{ then } a=10$$

$$\text{put } r = \frac{1}{2} \text{ in } (1), \text{ then } a=40$$

$\therefore$ The numbers are 10, 20, 40 or 40, 20, 10.

$\therefore$ Sum of the numbers = 70

Properties of G.P:-

- a, b, c are in G.P $\Leftrightarrow b^2 = ac$
- In a finite G.P, the product of the terms equidistant from the beginning and end is always same and is equal to the product of the first and last terms

(i.e) $a_2 a_{n-1} = a_3 a_{n-2} = a_4 a_{n-3} \dots = a_1 a_n$

- $a_1 \cdot a_2 \cdot a_3 \dots a_n = (\text{middle term})^n$, if n is odd
 $= (\text{Product of two middle terms})^{n/2}$, if n is even

- If $a_1, a_2, a_3, \dots, a_n$ are in G.P

a) $a_n, a_{n-1}, a_{n-2}, \dots, a_1$ are in G.P

b) $\lambda a_1, \lambda a_2, \lambda a_3, \dots, \lambda a_n$ are in G.P

$$(\lambda \in R - \{0\})$$

c) $a_1^n, a_2^n, a_3^n, \dots, a_n^n$ are in G.P for $n \in R$

d) $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in G.P

- If $a_1, a_2, a_3, \dots, a_n$ is a G.P of non zero, non negative terms then $\log a_1, \log a_2, \log a_3, \dots, \log a_n$ are in A.P and vice versa

Some facts about G.P :-

- If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are two G.P's with common ratio r_1 and r_2 respectively, then

a) $a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3, \dots, a_n \pm b_n$ are not in G.P

b) $a_1 b_1, a_2 b_2, a_3 b_3, \dots, a_n b_n$ are in G.P with common ratio $r_1 r_2$

c) $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots, \frac{a_n}{b_n}$ are in G.P with common ratio $\frac{r_1}{r_2}$

Increasing and decreasing G.P:-

- Let $a, ar, ar^2, \dots$ be G.P
- a) If $a > 0; r > 1$ then it is an increasing G.P
- b) If $a > 0; 0 < r < 1$ then it is decreasing G.P
- c) If $a < 0; r > 1$ then it is decreasing G.P
- d) If $a < 0; 0 < r < 1$ then it is an increasing G.P

W.E-18: If the sides of a triangle are in G.P and its larger angle is twice the smallest, then find the common ratio r satisfies the inequality.

Sol: Let the sides of a triangle be $a/r, a$ and ar , with $a > 0$ and $r > 1$. let α be the smallest angle. So that the largest angle is 2α . then α is opposite to the side a/r , and 2α is opposite to the side ar . Applying

sine rule, we get $\frac{a/r}{\sin \alpha} = \frac{ar}{\sin 2\alpha}$

$$\Rightarrow \frac{\sin 2\alpha}{\sin \alpha} = r^2 \Rightarrow r^2 = 2 \cos \alpha < 2$$

$$\Rightarrow r^2 < 2 \Rightarrow r < \sqrt{2}$$

$$\therefore 1 < r < \sqrt{2}$$

Geometric mean (G.M):- The geometric mean G of any two numbers 'a' and 'b' is given by $\sqrt{ab}$ where a, b are in G.P

- If $a_1, a_2, a_3, \dots, a_n$ be n numbers then geometric mean of these numbers is $(a_1 \cdot a_2 \cdot a_3 \dots a_n)^{\frac{1}{n}}$

- The n numbers $G_1, G_2, G_3, \dots, G_n$ are said to be geometric means between 'a' and 'b'. If $a, G_1, G_2, G_3, \dots, G_n, b$ are in G.P

Here a = First term ; $b = (n+2)th$ term

$$\text{then } r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}; G_1 = a \left(\frac{b}{a}\right)^{\frac{1}{n+1}};$$

$$G_2 = a \left(\frac{b}{a}\right)^{\frac{2}{n+1}} \dots; G_n = a \left(\frac{b}{a}\right)^{\frac{n}{n+1}}$$

$$G_1 G_2 G_3 \dots G_n = (\sqrt[n+1]{ab})^n = (GM \text{ of } a, b)^n$$

- If 'a' and 'b' are two numbers of opposite signs, the G.M. between them does not exist.

W.E-19: Find the geometric mean between -9 and -16.

Sol:

$$\text{Required G.M} = \sqrt{-9 \times -16} = (3i)(4i) = -12$$

W.E-20: If we insert two numbers between 3 and 81 so that the resulting sequence is G.P then find the numbers.

Sol: Let the two numbers be a and b, then 3, a, b, 81 are in G.P.

$$\therefore \text{nth term } T_n = AR^{n-1}; 81 = 3R^{4-1}$$

$$\Rightarrow R^3 = \frac{81}{3} = 27 \Rightarrow R^3 = 3^3 \Rightarrow R = 3$$

$$\therefore a = AR = 3 \times 3 = 9, b = AR^2 = 3 \times 3^2 = 27$$

Arithmetico - Geometric progression

(A.G.P): A sequence is called an arithmetico-geometric progression, if each term is the product of the corresponding terms of an A.P. and a G.P.,

➤ If a, a+d, a+2d, a+3d, is an A.P and b, br, br², is in G.P. then ab, (a+d)br, (a+2d)br², is an A.G.P

➤ The general form of an A.G.P is a, (a+d)r, (a+2d)r²,

General term of A.G.P:

General term of an A.G.P is

$$T_n = [a + (n-1)d] \cdot r^{n-1} \text{ where } a = \text{first term, } d = \text{common difference and } r = \text{common ratio.}$$

Sum to n term of an A.G.P :

$$S_n = \begin{cases} \frac{a}{1-r} + dr \frac{(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r} & (r \neq 1) \\ \frac{n}{2} [2a + (n-1)d] & (\text{when } r = 1) \end{cases}$$

➤ If the number of terms are infinite, then the sum of

$$\text{A.G.P is } S_\infty = \frac{a}{(1-r)} + \frac{dr}{(1-r)^2} \quad (\text{when } |r| < 1)$$

W.E-21: Find the nth term of arithmetico-geometric series $1-3x+5x^2-7x^3+\dots$

Sol: The given arithmetico-geometric series is $1-3x+5x^2-7x^3+\dots$. The A.P. corresponding to this series is 1, 3, 5, 7, and the G.P.

corresponding to this series is $1, (-x), (-x)^2, (-x)^3, \dots$ clearly, the nth term of the A.P. = $\{1 + (n-1)(2)\} = 2n-1$

and the nth term of G.P. = $\{1(-x)^{n-1}\} = (-1)^{n-1} \cdot x^{n-1}$

$\therefore$ the nth term of the given series

$$= (2n-1)(-1)^{n-1} \cdot x^{n-1} = (-1)^{n-1} (2n-1) \cdot x^{n-1}$$

To Find nth term by Difference Method:

If $T_1, T_2, \dots, T_n$ are terms of any series and their difference $(T_2 - T_1), (T_3 - T_2), (T_4 - T_3), \dots,$

$(T_n - T_{n-1})$ are either in A.P. or in G.P., then T_n and S_n of series may be found by the method of differences. Let $S_n = T_1 + T_2 + \dots + T_n$

$$\text{again } S_n = T_1 + T_2 + \dots + T_{n-1} + T_n$$

$$S_n - S_{n-1} = T_1 + (T_2 - T_1) + (T_3 - T_2) + \dots + (T_n - T_{n-1}) - T_n$$

$$T_n = T_1 + (T_2 - T_1) + (T_3 - T_2) + \dots + (T_n - T_{n-1})$$

$$T_n = T_1 + t_1 + t_2 + \dots + t_{n-1}$$

where $t_1, t_2, \dots, t_{n-1}$ are terms of the new series.

W.E-22: Find the sum of upto n terms of series : $5+7+13+31+85+\dots$

Sol: The difference between the successive terms are **2, 6, 18, 54,** Clearly it is a G.P. Let T_n be the nth term of the given series and S_n be the sum of its n terms, then

$$S_n = 5 + 7 + 13 + 31 + \dots + T_n \rightarrow (1)$$

$$S_n = 5 + 7 + 13 + \dots + T_{n-1} + T_n \rightarrow (2)$$

Subtracting (2) from (1)

$$0 = 5 + [2 + 6 + 18 + 54 + \dots + (T_n - T_{n-1})] - T_n$$

$$\Rightarrow 0 = 5 + 2 \frac{3^{n-1} - 1}{3 - 1} - T_n \Rightarrow T_n = 5 + (3^{n-1} - 1) = 4 + 3^{n-1}$$

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (4 + 3^{k-1}) = \sum_{k=1}^n 4 + \sum_{k=1}^n 3^{k-1}$$

$$= 4n + (1 + 3 + 3^2 + \dots + 3^{n-1}) = 4n + 1 \left(\frac{3^n - 1}{3 - 1} \right)$$

$$= 4n + \left(\frac{3^n - 1}{2} \right) = \frac{1}{2} (3^n + 8n - 1)$$

W.E-23: Find the Sum to infinity of the series

$$1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \dots \text{ is [AIEEE 2009]}$$

Sol: Let $S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \dots$ ---(1)

$$\frac{1}{3}S = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \dots$$
 ---(2)

Subtracting (2) from (1)

$$S\left(1 - \frac{1}{3}\right) = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \dots$$

$$S\left(\frac{2}{3}\right) = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$\frac{2}{3}S = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots = \frac{4}{3} + \frac{4}{3^2} \cdot \frac{3}{2} = 2 \quad \therefore S=3$$

W.E-24: Find the sum of the infinite series

$$1 + \frac{4}{3} + \frac{9}{3^2} + \frac{16}{3^3} + \dots \infty$$

Sol: This is clearly not an A.G.P Series, since 1, 4, 9, 16, ... are not in A.P. However their successive differences 4-1=3, 9-4=5, 16-9=7, ... are in A.P.

Let $S_{\infty} = 1 + \frac{4}{3} + \frac{9}{3^2} + \frac{16}{3^3} + \dots \infty$ ---(1)

$$\frac{1}{3}S_{\infty} = \frac{1}{3} + \frac{4}{3^2} + \frac{9}{3^3} + \dots \infty$$
 ---(2)

Subtracting (2) from (1)

$$\frac{2}{3}S_{\infty} = 1 + \frac{3}{3} + \frac{5}{3^2} + \frac{7}{3^3} + \dots \infty$$

$$\frac{1}{3} \cdot \frac{2}{3}S_{\infty} = \frac{1}{3} + \frac{3}{3^2} + \frac{5}{3^3} + \dots \infty$$

on Subtracting $\left(\frac{4}{9}\right) \cdot S_{\infty} = 1 + \frac{2}{3} + \frac{2}{3^2} + \dots \infty$

$$= 1 + \frac{2}{3} \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots \infty\right)$$

$$= 1 + \frac{2}{3} \left(\frac{1}{1 - \frac{1}{3}}\right) = 2 \quad \therefore S_{\infty} = \left(2 \times \frac{9}{4}\right) = \frac{9}{2}$$

Harmonic Progression (H.P): A sequence is in H.P, if the reciprocals of its terms form an A.P.

➤ In general H.P is of the form

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots, \frac{1}{a+(n-1)d}$$

where a = first term, d = common difference in A.P.

Properties of H.P:

➤ a, b, c are in H.P $\Leftrightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

➤ If $a_1, a_2, a_3, \dots, a_n$ are in H.P then

(a) $a_n, a_{n-1}, \dots, a_3, a_2, a_1$ are in H.P

(b) $\lambda a_1, \lambda a_2, \lambda a_3, \dots, \lambda a_n$ are in H.P ($\lambda \in R$)

(c) $\frac{a_1}{\lambda}, \frac{a_2}{\lambda}, \frac{a_3}{\lambda}, \dots, \frac{a_n}{\lambda}$ are in H.P where $\lambda \neq 0$

(d) If a, b are the first two terms of an H.P, then the

$$n^{\text{th}} \text{ term} = \frac{ab}{b + (n-1)(a-b)}$$

(e) If m^{th} term of H.P is 'n' and n^{th} term of H.P is 'm',

$$\text{then } T_r = \frac{mn}{r}$$

W.E.25: The 5th and 11th terms of an H.P are $\frac{1}{45}$

and $\frac{1}{69}$ respectively, then find 16th term.

Sol: The 5th and 11th terms of the corresponding A.P. are 45 and 69 respectively. Let a be the first term and d be the common difference of the corresponding A.P then, 5th term = $a + 4d = 45$ (i) and 11th term = $a + 10d = 69$ (ii) solving equations (i) and (ii), we get $a = 29, d = 4$ $\therefore$ the 16th term of the A.P = $a + 15d = 29 + 15(4) = 89$ hence, the 16th term of the H.P = $1/89$

Harmonic Mean (H.M):- The harmonic mean H of any two numbers a and b is given by

$$H = \frac{2ab}{a+b}, \text{ where } a, H, b \text{ are in H.P.}$$

➤ If $a_1, a_2, a_3, \dots, a_n$ be n numbers then H.M of

$$\text{these numbers is } H = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}$$

$$\Rightarrow \frac{1}{H} = \frac{1}{n} \left[\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right]$$

- The n numbers $H_1, H_2, H_3, \dots, H_n$ are said to be harmonic means between a and b if $a, H_1, H_2, H_3, \dots, H_n, b$ are in HP.

Here a = first term ; b = $(n+2)$ th term

If D is common difference of AP

$$\text{then } D = \frac{a-b}{(n+1)ab}; \quad \frac{1}{H_1} = \frac{1}{a} + \frac{a-b}{(n+1)ab};$$

$$\frac{1}{H_2} = \frac{1}{a} + \frac{2(a-b)}{(n+1)ab}, \dots, \frac{1}{H_n} = \frac{1}{a} + \frac{n(a-b)}{(n+1)ab}$$

$$\frac{1}{H_1} + \frac{1}{H_2} + \frac{1}{H_3} + \dots + \frac{1}{H_n} = \frac{n}{2} \left[\frac{1}{a} + \frac{1}{b} \right]$$

- If $x_1, x_2, x_3, \dots, x_n$ are n -H.M.'s between a and b ,

$$\text{then } x_1 = \frac{ab(n+1)}{b(n+1) + (a-b)},$$

$$x_2 = \frac{ab(n+1)}{b(n+1) + 2(a-b)}, \dots, x_n = \frac{ab(n+1)}{b(n+1) + n(a-b)}$$

W.E-26: Find two H.M's between $1/2, 4/17$.

Sol: Let x_1 and x_2 be two H.M's between $1/2, 4/17$

$$\therefore a = \frac{1}{2}, \quad b = \frac{4}{17}, \quad n = 2$$

$$x_1 = \frac{ab(2+1)}{b(2+1) + 1(a-b)} = \frac{3ab}{a+2b} = \frac{3\left(\frac{1}{2}\right)\left(\frac{4}{17}\right)}{\left(\frac{1}{2}\right) + 2\left(\frac{4}{17}\right)} = \frac{4}{11}$$

$$x_2 = \frac{ab(2+1)}{b(2+1) + 2(a-b)} = \frac{3ab}{2a+b} = \frac{3\left(\frac{1}{2}\right)\left(\frac{4}{17}\right)}{2\left(\frac{1}{2}\right) + \left(\frac{4}{17}\right)} = \frac{2}{7}$$

Relations between A.M, G.M, H.M:- Let A, G, H be A.M, G.M and H.M between two numbers a and b then

- $A = \frac{a+b}{2}; G = \sqrt{ab}; H = \frac{2ab}{a+b}$
 ➤ $A \geq G \geq H$
 ➤ A, G, H are in GP (i.e) $G^2 = AH$

- The equation having a and b as its roots is $x^2 - 2Ax + G^2 = 0$
 ➤ If A, G, H are A.M, G.M, H.M between three numbers a, b, c then the equation having a, b, c as its roots is $x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$
 ➤ $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the A.M, G.M & H.M between a and b for $n = 1, \frac{1}{2}, 0$ respectively
 ➤ If A and G be the A.M. and G.M between two positive numbers, then the numbers are $A \pm \sqrt{A^2 - G^2}$
 ➤ If the A.M. and G.M. between two numbers are in the ratio $m : n$, then the numbers are in the ratio $m + \sqrt{m^2 - n^2} : m - \sqrt{m^2 - n^2}$

W.E-27: Let two numbers have arithmetic mean 9 and geometric mean 4. then find the numbers are the roots of the quadratic equation.

Sol: The A.M. of the two numbers is $A=9$ and the G.M of two numbers is $G=4$

The quadratic equation whose roots are the numbers having A.M and G.M. are A, G respectively is $x^2 - 2Ax + G^2 = 0$. So, the required quadratic equation is $x^2 - 18x + 16 = 0$

W.E-28: Find two numbers whose arithmetic mean is 34 and geometric mean is 16.

Sol: Let the two numbers be a and b then $\frac{a+b}{2} = 34$

$$\text{and } \sqrt{ab} = 16$$

$$\Rightarrow a+b=68 \text{ and } ab=256$$

$$\therefore (a-b)^2 = (a+b)^2 - 4ab$$

$$= (68)^2 - 4(256) = 3600 \Rightarrow a-b=60$$

on solving $a+b=68$ and $a-b=60$, we get $a=64$, and $b=4$. thus, the required numbers are 64 and 4.

W.E-29: The H.M. between two numbers is $16/5$, their A.M. is A and G.M. is G . If $2A+G^2=26$ then find the numbers.

Sol: Given H.M of a and b is $\frac{2ab}{a+b} = \frac{16}{5}$

$$\Rightarrow a+b = \frac{5ab}{8} \rightarrow (1)$$

$$\text{Given } 2A+G^2=26 \Rightarrow 2\left(\frac{a+b}{2}\right)+ab=26$$

$$\Rightarrow (a+b)+ab=26 \Rightarrow \frac{5ab}{8}+ab=26 \Rightarrow ab=16$$

$$\text{From (1), } a+b = \frac{5}{8}(16) \Rightarrow a+b=10 \rightarrow (2)$$

$$\therefore (a-b)^2 = (a+b)^2 - 4ab = 100 - 64 = 36$$

$$\therefore (a-b)=6 \rightarrow (3) \text{ Solving (2) and (3)}$$

$$\therefore a=8, b=2$$

Weighted Means: Let $a_1, a_2, \dots, a_n$ be n positive real numbers and $m_1, m_2, m_3, \dots, m_n$ be n positive rational numbers. Then we have weighted Arithmetic mean A , Weighted geometric mean G and weighted harmonic mean H as

$$A = \frac{m_1 a_1 + m_2 a_2 + \dots + m_n a_n}{m_1 + m_2 + m_3 + \dots + m_n},$$

$$G = \left(a_1^{m_1} a_2^{m_2} \dots a_n^{m_n} \right)^{\frac{1}{m_1+m_2+\dots+m_n}} \quad \text{and}$$

$$H = \frac{m_1 + m_2 + \dots + m_n}{\frac{m_1}{a_1} + \frac{m_2}{a_2} + \dots + \frac{m_n}{a_n}} \quad \text{Then we have}$$

$A \geq G \geq H$. Moreover equality hold at either place $\Leftrightarrow a_1 = a_2 = \dots = a_n$

W.E-30: If $2p+3q+4r=15$, then find the maximum value of $p^3 q^5 r^7$.

Sol: Since

$$\frac{2p}{3} + \frac{2p}{3} + \frac{2p}{3} + \left(\frac{3q}{5} + \dots + \frac{3q}{5} \right) (5 \text{ times}) + \left(\frac{4r}{7} + \dots + \frac{4r}{7} \right) (7 \text{ times})$$

$$\geq 15 \sqrt[15]{\left(\frac{2p}{3} \right)^3 \left(\frac{3q}{5} \right)^5 \left(\frac{4r}{7} \right)^7} \quad (\because AM \geq GM)$$

$$p^3 q^5 r^7 \frac{2^3 3^5 4^7}{3^3 5^5 7^7} \leq 1 \Rightarrow p^3 q^5 r^7 \leq \frac{5^5 7^7}{2^3 3^2 4^7}$$

Summation of some series of natural numbers:

$$\Rightarrow \sum_{k=1}^n 1 = 1+1+\dots+1(n \text{ terms}) = n$$

$$\Rightarrow \sum_{k=1}^n k = 1+2+3+\dots+n = \frac{1}{2}n(n+1)$$

$$\Rightarrow \sum_{k=1}^n k^2 = 1^2+2^2+\dots+n^2 = \frac{1}{6}n(n+1)(2n+1)$$

$$\begin{aligned} \Rightarrow \sum_{k=1}^n k^3 &= 1^3+2^3+\dots+n^3 = \left(\sum_{k=1}^n k \right)^2 = (1+2+3+\dots+n)^2 \\ &= \left[\frac{1}{2}n(n+1) \right]^2 = \frac{1}{4}n^2(n+1)^2 \end{aligned}$$

$$\Rightarrow 1+3+5+\dots n \text{ terms} = \sum_{k=1}^n (2k-1) = n^2$$

$$\Rightarrow 2+4+6+\dots n \text{ terms} = \sum_{k=1}^n (2k) = n(n+1)$$

$$\begin{aligned} \Rightarrow 1^2+3^2+5^2+\dots n \text{ terms} &= \sum_{k=1}^n (2k-1)^2 \\ &= \frac{n}{3}(4n^2-1) \end{aligned}$$

$$\begin{aligned} \Rightarrow 2^2+4^2+6^2+\dots n \text{ terms} &= \sum_{k=1}^n (2k)^2 \\ &= \frac{2}{3}n(n+1)(2n+1) \end{aligned}$$

$$\begin{aligned} \Rightarrow 1^3+3^3+5^3+\dots n \text{ terms} &= \sum_{k=1}^n (2k-1)^3 \\ &= n^2(2n^2-1) \end{aligned}$$

$\Rightarrow$ Sum of n terms of series

$$1^2-2^2+3^2-4^2+5^2-\dots = \begin{cases} \frac{n(n+1)}{2}, & \text{if } n \text{ is odd} \\ \frac{-n(n+1)}{2}, & \text{if } n \text{ is even} \end{cases}$$

W.E-31: $1^3-2^3+3^3-4^3+\dots+9^3 =$

$$\begin{aligned} \text{Sol: } 1^3-2^3+3^3-4^3+\dots+9^3 &= \\ &= (1^3+2^3+\dots+9^3) - 2(2^3+4^3+\dots+8^3) \end{aligned}$$

$$\begin{aligned} &= \left[\frac{9(9+1)}{2} \right]^2 - 2 \times 2^3(1^3+2^3+3^3+4^3) \\ &= 2025 - 1600 = 425 \end{aligned}$$

Note: If $|x| < 1$ then

1. $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$
2. $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$
3. $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots$
4. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$
5. $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$
6. $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots$

General rule for finding the values of recurring decimal: Let X denote the figure which do not recur and assume they are l in number.

Let Y denote recurring period of consisting of m figures. Let R denote the value of recurring decimal

then $R = XYYY\dots$ (or) $R = X\dot{Y}$

$$\therefore 10^l R = X.YYY\dots \text{ and } 10^{l+m} R = XY.YYY\dots$$

$$\therefore \text{ Subtracting we get } R = \frac{XY - X}{10^{l+m} - 10^l}$$

E.g: $0.6\dot{2}\dot{3} = \frac{623 - 6}{990} = \frac{617}{990}$

E.g: $1.2\dot{4}\dot{3} = 1 + \frac{243 - 2}{990} = 1 + \frac{241}{990} = \frac{1231}{990}$

Sum of the products of two terms of a sequence :

To obtain the sum $\sum_{i < j} a_i a_j$, we use the identity

$$2 \sum_{i < j} a_i a_j = (a_1 + a_2 + \dots + a_n)^2 - (a_1^2 + a_2^2 + \dots + a_n^2)$$

Cauchy-Schwartz's Inequality If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are $2n$ real numbers, then

$$(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)$$

$$(b_1^2 + b_2^2 + \dots + b_n^2) \text{ with the equality holding if and}$$

$$\text{only if } \frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}$$



C.U.Q

1. If $a_1, a_2, a_3, \dots, a_{n+1}$ are in A.P., of non zero

terms then $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_n a_{n+1}}$, is

- 1) $\frac{n-1}{a_1 a_{n+1}}$ 2) $\frac{1}{a_1 a_{n+1}}$ 3) $\frac{n+1}{a_1 a_{n+1}}$ 4) $\frac{n}{a_1 a_{n+1}}$

2. If $a_1, a_2, a_3, \dots, a_n$ is an A.P of non zero terms such that

$$\frac{1}{a_1 a_n} + \frac{1}{a_2 a_{n-1}} + \frac{1}{a_3 a_{n-2}} + \dots + \frac{1}{a_n a_1} = \lambda \left\{ \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right\} \text{ then } \lambda =$$

- 1) 2 2) $a_1 + a_n$ 3) $2(a_1 + a_n)$ 4) $\frac{2}{a_1 + a_n}$

3. If $A_1, A_2, A_3, \dots, A_n$ are 'n' numbers inserted between a, b to form A.P. then $A_1 + A_2 + A_3 + \dots + A_n$

- 1) $\frac{a+b}{2}$ 2) $\frac{n}{2}(a+b)$ 3) $\frac{n}{4}(a+b)$ 4) $\frac{n}{3}(a+b)$

4. If the arithmetic mean between a and b is

$$\frac{a^{n+1} + b^{n+1}}{a^n + b^n}, \text{ then } n =$$

- 1) 0 2) 1 3) -1 4) 1/2

5. If the number of terms in a G.P. is odd, then product of terms =

- 1) (middle term)^{no of terms}
2) (middle term)^{no of terms-1}
3) (middle term)^{no of even terms+2}
4) (middle term)^{no of terms+1}

6. If the number of terms in a G.P. is even, then product of terms =

- 1) (G.M. of middle terms)^{no of terms}
2) (A.M. of middle terms)^{no of terms}
3) (H.M. of middle terms)^{no of terms}
4) (G.M. of middle terms)^{no of terms-1}

7. If $G_1, G_2, G_3, \dots, G_n$ are n numbers inserted between a, b of G.P. then common ratio is

1) $\left(\frac{a}{b}\right)^{\frac{1}{n+1}}$ 2) $\left(\frac{b}{a}\right)^{\frac{1}{n+1}}$ 3) $\left(\frac{b}{a}\right)^{n+1}$ 4) $\left(\frac{a}{b}\right)^{n+1}$

8. If $G_1, G_2, G_3, \dots, G_n$ are n numbers inserted between a, b to form a G.P. then $G_1 G_2 G_3 \dots G_n =$

1) $\sqrt{ab}$ 2) $(\sqrt{ab})^n$ 3) $(\sqrt{ab})^{n+1}$ 4) $\left(\frac{a+b}{2}\right)^n$

9. If the geometric mean between a and b is

$\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ then $n =$

1) 0 2) $1/2$ 3) $-1/2$ 4) $1/4$

10. If the harmonic mean between a and b is

$\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$, then $n =$

1) 0 2) -1 3) $-1/2$ 4) 1

C.U.Q KEY

1) 4 2) 4 3) 2 4) 1 5) 1
6) 1 7) 2 8) 2 9) 3 10) 2



LEVEL-I (C.W)

1. In an A.P., if first term is 4, 9th term is 20, then 15th term is

1) 16 2) 32 3) 18 4) 36

2. The number of numbers that are divisible by 9 between 1 and 1000 is

1) 101 2) 110 3) 111 4) 100

3. Let T_r be the r^{th} term of an A.P. whose first term is a and common difference is d . If for some positive integers m, n ,

$m \neq n, T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then $a-d =$

1) 0 2) $\frac{1}{m} + \frac{1}{n}$ 3) $\frac{1}{mn}$ 4) 1

4. If p^{th}, q^{th}, r^{th} terms of an A.P are a, b, c then $a(q-r) + b(r-p) + c(p-q) =$

1) 0 2) 1 3) $a + b + c$ 4) abc

5. If the numbers a, b, c, d, e form an A.P., then the value of $a - 4b + 6c - 4d + e$ is
1) 1 2) 2 3) 0 4) None

6. If $S_n = nP + \frac{n}{2}(n-1)Q$, where S_n denotes the sum of the first n terms of an A.P., then the common difference is

1) $P+Q$ 2) $2P+3Q$ 3) $2Q$ 4) Q

7. In an A.P., if common difference is 2, sum to n terms is 49, 7th term is 13, then $n =$
1) 0 2) 5 3) 7 4) 13

8. Consider an A.P. with first term a and common difference d . Let S_k denote the

sum of the first k terms. If $\frac{S_{kx}}{S_x}$ is independent of x , then

1) $a = 2d$ 2) $a = d$ 3) $2a = d$ 4) $2a = 3d$

9. If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \dots + a_{23} + a_{24} =$
1) 909 2) 75 3) 750 4) 900

10. The degree of the expression $(1+x)(1+x^6)(1+x^{11}) \dots (1+x^{101})$ is
1) 1081 2) 1061 3) 1071 4) 1091

11. If the sum of three numbers which are in A.P is 27 and the product of first and last is 77, then the numbers are

1) 7, 9, 11 2) 6, 9, 12
3) 7, 10, 11 4) 7, 11, 9

12. If the ratio of n^{th} terms of two A.P.'s is $(2n+8):(5n-3)$, then the ratio of the sums of their n terms is

1) $(2n+18):(5n+1)$ 2) $(5n-1):(2n+18)$
3) $(2n+18):(5n-1)$ 4) $(3n+18):(4n+1)$

13. Between 1 and 31 are inserted m arithmetic means, so that the ratio of the 7th and $(m-1)^{th}$ means is 5:9. Then the value of m is

1) 12 2) 13 3) 14 4) 15

14. If $S_1, S_2, S_3, \dots$ are the sums of first n natural numbers, their squares and their cubes respectively, then $S_3(1+8S_1) =$

1) S_2^2 2) $9S_2$ 3) $9S_2^2$ 4) $3S_2^2$

15. If 6th term of a G.P. is $-1/32$ and 9th term is $1/256$, then 11th term =
1) 1024 2) $1/1024$ 3) $1/256$ 4) $1/512$
16. If $(1-y)(1+2x+4x^2+8x^3+16x^4+32x^5)=1-y^6$, ($y \neq 1$), then a value of y/x is
1) $1/2$ 2) 2 3) $25/24$ 4) $24/25$
17. If the sum of three numbers in a G.P. is 26 and the sum of products taken two at a time is 156, then the numbers are
1) 2, 6, 18 2) 1, 8, 64 3) 1, 5, 25 4) 1, 4, 1
18. One of the 5 geometric means between $\frac{1}{3}$ and 243 is
1) 79 2) 80 3) 81 4) 82
19. If the fourth term of a H.P. is $1/3$ and 7th term is $1/4$, then 16th term is
1) $1/5$ 2) $1/6$ 3) $1/7$ 4) $1/8$
20. If a, b, c are in H.P. and $ab + bc + ca = 15$, then $ca =$
1) 5 2) 7 3) 9 4) 10
21. If $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$ upto $\infty = \frac{\pi^2}{6}$, then value of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$ up to ∞ is
1) $\frac{\pi^2}{4}$ 2) $\frac{\pi^2}{6}$ 3) $\frac{\pi^2}{8}$ 4) $\frac{\pi^2}{12}$
22. The sum of $\frac{3}{1.2} \cdot \frac{1}{2} + \frac{4}{2.3} \cdot \left(\frac{1}{2}\right)^2 + \frac{5}{3.4} \cdot \left(\frac{1}{2}\right)^3 + \dots$ to n terms is equal to
1) $1 - \frac{1}{(n+1)2^n}$ 2) $1 - \frac{1}{n.2^{n-1}}$
3) $1 + \frac{1}{(n+1).2^n}$ 4) $1 + \frac{1}{n.2^{n-1}}$
23. If $\sum_{r=1}^n (r^2 + 1)r! = 200 \times 201!$, then n =
1) 200 2) 201 3) 199 4) 202

LEVEL-I (C.W)-KEY

- 1) 2 2) 3 3) 1 4) 1 5) 3 6) 4
7) 3 8) 3 9) 4 10) 3 11) 1 12) 3
13) 3 14) 3 15) 2 16) 2 17) 1 18) 3
19) 3 20) 1 21) 3 22) 1 23) 1

LEVEL-I (C.W)-HINTS

1. $t_{15} = a + 14d$
2. Numbers div. by 9 are 9, 18, 27, ..., 999. It is an A.P. $t_n = 999$, find n
3. $T_m = a + (m-1)d = \frac{1}{n}$, $T_n = a + (n-1)d = \frac{1}{m}$
 $T_m - T_n = \frac{1}{n} - \frac{1}{m}$, find $d = \frac{1}{mn}$, using T_m , find a
- 4.. $a = A + (p-1)D$, $b = A + (q-1)D$, $c = A + (r-1)D$ find a-b, b-c, c-a
- 5.. Let D be the common difference of the A.P. Then,
 $a - 4b + 6c - 4d + e$
 $= a - 4(a+D) + 6(a+2D) - 4(a+3D) + (a+4D) = 0$
6. We have $S_n = nP + \frac{n}{2}(n-1)Q$
compare with $S_n = \frac{n}{2}[2a + (n-1)d]$
7. $d = 2$ and $T_7 = 13$, find a, use in S_n
8. We have $\frac{S_{kx}}{S_x} = \frac{\frac{kx}{2}\{2a + (kx-1)d\}}{\frac{x}{2}\{2a + (x-1)d\}}$
 $\Rightarrow \frac{S_{kx}}{S_x} = \frac{k[2a + (kx-1)d]}{[2a + (x-1)d]} = \frac{k[(2a-d) + kxd]}{[(2a-d) + xd]}$
Clearly, $\frac{S_{kx}}{S_x}$ is independent of x, if $d = 2a$
9. $(a_1 + a_{24}) + (a_5 + a_{20}) + (a_{10} + a_{15}) = 225$
 $3(a_1 + a_{24}) = 225 \Rightarrow (a_1 + a_{24}) = 75$
and $a_1 + a_2 + \dots + a_{24} = \frac{n}{2}(a_1 + a_{24})$
10. Degree of the expression is $1+6+11+\dots+101$
 $101 = 1 + 5(n-1) \Rightarrow n = 21$
 $\therefore \text{Degree} = \frac{21}{2}(1+101) = 1071$
11. $3a = 27$ and $a^2 - d^2 = 77$

12. Required ratio (replace n with $\frac{n+1}{2}$)

$$= \left\{ \frac{2(n+1)}{2} + 8 \right\} : \left\{ \frac{5(n+1)}{2} - 3 \right\}$$

13. $31 = 1 + (m+1)d$ & $\frac{1+7d}{1+(m-1)d} = \frac{5}{9}$

14. $S_1 = \frac{n(n+1)}{2}, S_2 = \frac{n(n+1)(2n+1)}{6}$

$$\text{and } S_3 = \frac{n^2(n+1)^2}{4}$$

15. find $\frac{t_9}{t_6}$

16. $1 - y^6 = (1-y)(1+y+y^2+y^3+y^4+y^5)$

17. $a + ar + ar^2 = 26; a \cdot ar + a \cdot ar^2 + ar \cdot ar^2 = 156$

18. $a = \frac{1}{3}, ar^5 = 243; \therefore r = 3$

19. $\frac{1}{a+3d} = \frac{1}{3}$ and $\frac{1}{a+6d} = \frac{1}{4}$

20. $b = \frac{2ac}{a+c}$

21. let $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = x;$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6}$$

Sum of odd terms + sum of even terms = $\frac{\pi^2}{6} + x$

$$\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) + \frac{1}{2^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \dots \right) = \frac{\pi^2}{6} + x$$

$$x + \frac{1}{4} \frac{\pi^2}{6} = \frac{\pi^2}{6}$$

22. $t_n = \frac{n+2}{n(n+1)} \left(\frac{1}{2} \right)^n = \frac{1}{n} \cdot \left(\frac{1}{2} \right)^{n-1} - \frac{1}{n+1} \left(\frac{1}{2} \right)^n$

23. $t_r = (r^2 + 1)r! = (r+1)^2 r! - 2r r!$
 $= (n+1)!(n+2-2) = n.(n+1)!$



LEVEL-I (H.W)

1. If the first term of an A.P is -1 and common difference is -3 , then 12th term is

1) 34 2) 32 3) -32 4) -34

2. If the sum to n terms of an A.P. is

$3n^2 + 5n$ while $T_m = 164$, then value of m is

1) 25 2) 26 3) 27 4) 28

3. Let T_r be the r th term of an AP for $r=1, 2, \dots$ If for some positive integers m and n we have $T_m = 1/n$ and $T_n = 1/m$, the $T_{mn} =$

1) $-1/mn$ 2) $1/m + 1/n$

3) 1 4) 0

4. The interior angles of a polygon are in A.P. If the smallest angle is 100° and the common difference is 4° , then the number of sides is

1) 5 2) 7 3) 36 4) 44

5. If a, b, c, d, e, f are in A.P., then $e-c$ is equal to

1) $2(c-a)$ 2) $2(d-c)$ 3) $f-e$ 4) $d-c$

6. If the ratio between the sums of n terms of two A.P.'s is $3n+8:7n+15$, then the ratio between their 12th terms is

1) $16:7$ 2) $7:16$ 3) $74:169$ 4) $169:74$

7. If the sum of the first ten terms of an A.P is four times the sum of its first five terms, then ratio of the first term to the common difference is

1) $1:2$ 2) $2:1$ 3) $1:4$ 4) $4:1$

8. If S_n denotes the sum of n terms of an A.P., then $S_{n+3} - 3S_{n+2} + 3S_{n+1} - S_n =$

1) 0 2) 1 3) 3 4) 2

9. In an A.P of 99 terms, the sum of all the odd numbered terms is 2550. Then the sum of all 99 terms is

1) 5039 2) 5029 3) 5019 4) 5049

10. If the first, second and the last terms of an A.P. are a, b, c respectively, then the sum of the A.P. is

1) $\frac{(a+b)(a+c-2b)}{2(b-a)}$ 2) $\frac{(b+c)(a+b-2c)}{2(b-a)}$

3) $\frac{(a+c)(b+c-2a)}{2(b-a)}$ 4) $\frac{(a+2c)(b+c+2c)}{2(b-a)}$

11. Four numbers are in arithmetic progression. The sum of first and last terms is 8 and the product of both middle terms is 15. The least number of the series is.

1) 4 2) 3 3) 2 4) 1

12. If n arithmetic means are inserted between 2 and 38, then the sum of the resulting series is obtained as 200, then the value of n is

1) 6 2) 8 3) 9 4) 10

13. If $m > 1$ and $n \in N$ then

1. $\frac{1^m + 2^m + \dots + n^m}{n} > \left(\frac{n+1}{2}\right)^m$

2. $\frac{1^m + 2^m + \dots + n^m}{n} < \left(\frac{n+1}{2}\right)^m$

3. $\frac{1^m + 2^m + \dots + n^m}{n} \geq 1$ 4. $\frac{1^m + 2^m + \dots + n^m}{n} \leq 1$

14. Sum of the series

$S = 1 + \frac{1}{2}(1+2) + \frac{1}{3}(1+2+3) + \frac{1}{4}(1+2+3+4) + \dots$ upto 20 terms is

1) 110 2) 111 3) 115 4) 116

15. The first and second terms of a G.P are x^{-4} and x^n respectively. If x^{52} is the eighth term of the same progression, then n is equal to

1) 13 2) 4 3) 5 4) 3

16. How many terms of the series $1+3+9+ \dots$ sum to 364?

1) 5 2) 6 3) 4 4) 3

17. If a , b and c are in G.P., then

$\frac{b-a}{b-c} + \frac{b+a}{b+c} =$

1) $b^2 - c^2$ 2) ac 3) ab 4) 0

18. If x , y , z are the three geometric means between 6, 54, then $z =$

1) $9\sqrt{3}$ 2) 18 3) $18\sqrt{3}$ 4) 27

19. H_1, H_2 are 2 H.M.'s between a , b then

$\frac{H_1 + H_2}{H_1 \cdot H_2} =$

1) $\frac{ab}{a+b}$ 2) $\frac{a+b}{ab}$ 3) $\frac{a-b}{ab}$ 4) $\frac{ab}{a-b}$

20. If $H_1, H_2, \dots, H_n$ are n harmonic means between a and b ($a \neq b$), then the value of

$\frac{H_1 + a}{H_1 - a} + \frac{H_n + b}{H_n - b} =$

1) $n+1$ 2) $n-1$ 3) $2n$ 4) $2n+3$

21. If $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$, then

$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$ is equal to

1) $\frac{\pi^2}{36}$ 2) $\frac{\pi^4}{48}$ 3) $\frac{\pi^2}{72}$ 4) $\frac{\pi^4}{96}$

22. The rational number which is equal to the number $2.\overline{357}$ with recurring decimal is

1) $\frac{2355}{1001}$ 2) $\frac{2370}{999}$ 3) $\frac{2355}{999}$ 4) $\frac{2359}{991}$

LEVEL-I (H.W)- KEY

1) 4 2) 3 3) 3 4) 1 5) 2 6) 2
7) 1 8) 1 9) 4 10) 3 11) 4 12) 2
13) 1 14) 3 15) 2 16) 2 17) 4 18) 3
19) 2 20) 3 21) 4 22) 3

LEVEL-I (H.W)-HINTS

- $t_{12} = a + (12-1)d$
- $T_m = S_m - S_{m-1}$
- $T_m = a + (m-1)d = \frac{1}{n}$, $T_n = a + (n-1)d = \frac{1}{m}$
 $T_m - T_n = \frac{1}{n} - \frac{1}{m}$, find $d = \frac{1}{mn}$, using T_m ,
find a and T_{mn}
- Sum of interior angles of a polygon of n sides
 $= (n-2) 180^\circ = \frac{n}{2}[2(100) + (n-1)4]$
- Let A be first term and D be c.d
 $e = A + 4D, c = A + 2D \therefore e - c = 2D$, check with option
- Ratio of the sums of n terms $= \frac{3n+8}{7n+15}$
 $\therefore$ Ratio of n^{th} terms Replace n with $(2n-1)$
 $= \frac{3(2n-1)+8}{7(2n-1)+15} = \frac{6n+5}{14n+8}$

$$\Rightarrow \text{Ratio of } 12^{\text{th}} \text{ terms} = \frac{6 \times 12 + 5}{14 \times 12 + 8} = \frac{77}{176} = \frac{7}{16}$$

$$7. S_{10} = 4 S_5$$

$$8. (S_{n+3} - S_{n+2}) - 2(S_{n+2} - S_{n+1}) + (S_{n+1} - S_n) = d - 2d + d = 0$$

$$9. \frac{50}{2}(a_1 + a_{99}) = 2550 \Rightarrow a_1 + a_{99} = 102$$

$$\text{sum of all the terms} = \frac{99}{2}[a_1 + a_{99}] = 5049$$

10. Let there be n terms in the A.P. Then,

$$c = a + (n-1)(b-a) \Rightarrow n = \frac{b+c-2a}{b-a}$$

$$\therefore \text{Sum of } n \text{ terms} = \frac{n}{2}(a+c) = \frac{(b+c-2a)(a+c)}{2(b-a)}$$

11. Take A.P. as $(a-3d)$, $a-d$, $a+d$, $a+3d$

12. Total no. of terms in A.P. is $n+2$

$$\text{given that } \left(\frac{n+2}{2}\right)(2+38) = 200$$

$$13. \frac{1^m + 2^m + \dots + n^m}{n} > \left(\frac{1+2+3+\dots+n}{n}\right)^m > \left(\frac{n+1}{2}\right)^m$$

$$14. S = \sum \frac{n(n+1)}{2 \cdot n} = \sum \left(\frac{n+1}{2}\right)$$

15. The common ratio of the G.P.'s x^{n+4}

$$\therefore x^{52} = \text{Eighth term}$$

$$\Rightarrow x^{52} = x^{-4} (x^{n+4})^7 \Rightarrow 7n = 28 \Rightarrow n = 4$$

$$16. \frac{1(3^n - 1)}{3 - 1} = 364 \text{ find } n$$

17. $b^2 = ac$ and simplifying the given

$$18. a = 6, ar^4 = 54 \Rightarrow r = \sqrt{3}$$

$$19. \frac{1}{H_1} + \frac{1}{H_2} + \dots + \frac{1}{H_n} = \frac{n}{2} \left(\frac{1}{a} + \frac{1}{b}\right)$$

$$20. \text{Use } \frac{1}{H_1} + \frac{1}{H_2} + \dots + \frac{1}{H_n} = \frac{n}{2} \left(\frac{1}{a} + \frac{1}{b}\right),$$

$$\text{find } \frac{a}{H_1} \text{ and } \frac{b}{H_n}$$

$$21. \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90},$$

$$\left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots\right) + \left(\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots\right) = \frac{\pi^4}{90}$$

$$\left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots\right) + \frac{1}{2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots\right) = \frac{\pi^4}{90}$$

$$\left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots\right) + \frac{1}{16} \left(\frac{\pi^4}{90}\right) = \frac{\pi^4}{90}, \text{ simplify}$$

22. let $x = 2.357357357\dots$

$$1000x = 2357.357357; \text{ subtract}$$

LEVEL-II (C.W)

1. If $a_r > 0$, $r \in N$ and $a_1, a_2, a_3, \dots, a_{2n}$ are A.P.

$$\text{then } \frac{a_1 + a_{2n}}{\sqrt{a_1} + \sqrt{a_{2n}}} + \frac{a_2 + a_{2n-1}}{\sqrt{a_2} + \sqrt{a_{2n-1}}} + \frac{a_3 + a_{2n-2}}{\sqrt{a_3} + \sqrt{a_{2n-2}}} + \dots + \frac{a_n + a_{n+1}}{\sqrt{a_n} + \sqrt{a_{n+1}}} =$$

$$1) n-1 \quad 2) \frac{n(a_1 + a_{2n})}{\sqrt{a_1} + \sqrt{a_{2n+1}}} \quad 3) \frac{n-1}{\sqrt{a_1} + \sqrt{a_{n+1}}} \quad 4) n+1$$

2. If the sum of the first $2n$ terms of the A.P. 2, 5, 8, is equal to the sum of the first n terms of the A.P. 57, 59, 61, then $n =$

$$1) 10 \quad 2) 12 \quad 3) 11 \quad 4) 13$$

3. Eleven A.M.'s are inserted between 28 and 10 then the number of integral A.M.'s

$$1) 5 \quad 2) 4 \quad 3) 6 \quad 4) 3$$

4. If t_n denotes n^{th} term of the series $2+3+6+11+18+\dots$ then t_{50}

$$1) 2+49^2 \quad 2) 2+48^2 \quad 3) 2+50^2 \quad 4) 2+51^2$$

5. The 10^{th} common term between series $3+7+11+\dots$ and $1+6+11+\dots$ is

$$1) 191 \quad 2) 190 \quad 3) 192 \quad 4) 194$$

6. Let $f(x)$ be a polynomial function of second degree. If $f(a) = f(-a)$ and a, b, c are in A.P. then $f'(a), f'(b), f'(c)$ are in

$$1) \text{ G.P. } \quad 2) \text{ H.P. } \quad 3) \text{ A.G.P. } \quad 4) \text{ A.P. }$$

7. In a G.P. of positive terms, for a fixed n , the n^{th} term is equal to sum of the next two terms. Then the common ratio of the G.P. is [AIEEE 2007]

$$1) 2 \cos 18^\circ \quad 2) \sin 18^\circ \\ 3) \cos 18^\circ \quad 4) 2 \sin 18^\circ$$

8. The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48. If the terms of the geometric progression are alternately positive and negative, then the first term is

[AIEEE - 2008]

- 1) 12 2) 4 3) -4 4) -12

9. If $a_i \in R - \{0\}, i = 1, 2, 3, 4$ and $x \in R$ and

$$\left(\sum_{i=1}^3 a_i^2 \right) x^2 - 2x \left(\sum_{i=1}^3 a_i a_{i+1} \right) + \sum_{i=2}^4 a_i^2 \geq 0, \text{ then}$$

a_1, a_2, a_3, a_4 are in

- 1) A.P 2) G.P 3) H.P 4) A.G.P

10. If $x \in \{1, 2, 3, \dots, 9\}$ and $f_n(x) = xxx\dots x$ (n digits), then $f_n^2(3) + f_n(2)$

- 1) $2f_{2n}(1)$ 2) $f_n^2(1)$
3) $f_{2n}(1)$ 4) $-f_{2n}(4)$

11. The sum of the series $\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \dots$ to n terms is

- 1) $n + \frac{1}{2}(3^n - 1)$ 2) $n - \frac{1}{2}(3^n - 1)$
3) $n - \frac{1}{2}(3^{-n} - 1)$ 4) $n - \frac{1}{2}(1 - 3^{-n})$

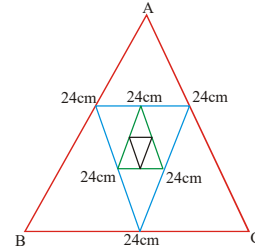
12. If $x = 1 + a + a^2 + \dots \infty$, where $|a| < 1$ and $y = 1 + b + b^2 + \dots \infty$ where $|b| < 1$, then $1 + ab + a^2 b^2 + \dots \infty =$

- 1) $\frac{xy}{x+y-1}$ 2) $\frac{x+y-1}{yx}$
3) $\frac{xy}{x+y+1}$ 4) $\frac{x+y+1}{xy}$

13. Consider an infinite geometric series with the first term a and common ratio r. If its sum is 4 and the second term is $3/4$, then
1) $a = 4/7, r = 3/7$ 2) $a = 2, r = 3/8$
3) $a = 3/2, r = 1/2$ 4) $a = 3, r = 1/4$

14. The sum of an infinite number of terms of a G.P. is 20, and the sum of their squares is 100, then the first term of the G.P. is
1) 5 2) $8/5$ 3) $3/5$ 4) 8

15. One side of an equilateral triangle is 24 cm. The midpoints of its sides are joined to form another triangle whose midpoints are in turn joined to form still another triangle this process continues indefinitely. The sum of the perimeters of all the triangles



- 1) 144 cm 2) 142 cm 3) 140 cm 4) 130 cm

16. Value of $y = (0.36)^{\log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \text{upto } \infty \right)}$ is

- 1) 0.9 2) 0.8 3) 0.6 4) 0.25

17. If the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$ are in G.P., then

- 1) $c^3 a = b^3 d$ 2) $ca^3 = bd^3$
3) $a^3 b = c^3 d$ 4) $ab^3 = cd^3$

18. The number a, b, c and d are in G.P. Then the value of

$$(a-c)^2 + (b-c)^2 + (b-d)^2 - (a-d)^2 \text{ is}$$

- 1) 0 2) 1 3) 2 4) 4

19. If a be one A.M and G_1 and G_2 be then geometric means between b and c then $G_1^3 + G_2^3 =$

- 1) abc 2) 2abc 3) 3abc 4) 4abc

20. If x, y, z are three real numbers of the same sign then the value of $x/y + y/z + z/x$ lies in the interval

- 1) $[2, \infty)$ 2) $[3, \infty)$ 3) $(3, \infty)$ 4) $(-\infty, 3)$

21. If a, b, c are the sides of a triangle then

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c}$$

- 1) ≤ 3 2) ≥ 3 3) ≥ 2 4) ≤ 2

22. The greatest value of $x^2 y^3$ where $3x + 4y = 5$ is

- 1) $\frac{3}{8}$ 2) $\frac{3}{16}$ 3) $\frac{6}{5}$ 4) $\frac{9}{5}$

23. If $a+b+c=6$ then
 $\sqrt{4a+1} + \sqrt{4b+1} + \sqrt{4c+1} =$
 1) 9 2) 8 3) 7 4) 6
24. If a, b, c be the p^{th} , q^{th} and r^{th} terms respectively of an A.P. and G.P. both, then the product of the roots of equation $a^b b^c c^a x^2 - abc x + a^c b^a c^b = 0$ is equal to
 1) -1 2) 1
 3) 2 4) $(b-c)(c-a)(a-b)$
25. If $p^a = q^b = r^c = s^d$, and p, q, r, s are in G.P. then a, b, c, d are in
 1) A.P 2) G.P 3) H.P. 4) A.G.P
26. If $a_1, a_2, a_3, \dots, a_n$ are in H.P then
 $a_1 a_2 + a_2 a_3 + a_3 a_4 + \dots + a_{n-1} a_n =$
 1) $(n-1)a_1 a_n$ 2) $(n+1)a_1 a_n$
 3) $(n-2)a_1 a_n$ 4) $(n+4)a_1 a_n$
27. If a, b, c are non - zero real numbers such that $3(a^2 + b^2 + c^2 + 1)$
 $= 2(a+b+c+ab+bc+ca)$, then a, b, c are in
 1) A.P. only 2) G.P. only
 3) A.P., G.P., and H.P 4) A.P. and G.P both
28. If a, b, c are in H.P., then the straight line
 $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$ always passes through a fixed point is
 1) $(-1, -2)$ 2) $(-1, 2)$
 3) $(1, -2)$ 4) $(1, -1/2)$
29. The harmonic mean of two numbers is 4. Their arithmetic mean is A and geometric mean is G. If G satisfies $2A + G^2 = 27$, then the numbers are
 1) 1, 13 2) 9, 12 3) 3, 6 4) 4, 8
30. If n is an odd integer greater than or equal to 1 then the value of
 $n^3 - (n-1)^3 + (n-2)^3 - \dots + (-1)^{n-1} \cdot 1^3$ is
 1) $\frac{(n+1)^2 \cdot (2n-1)}{4}$ 2) $\frac{(n-1)^2 \cdot (2n-1)}{4}$
 3) $\frac{(n+1)^2 \cdot (2n+1)}{4}$ 4) $\frac{(n+1)^2 \cdot (2n+1)}{8}$

31. If $a = \sum_{r=1}^{\infty} \frac{1}{r^4}$, then $\sum_{r=1}^{\infty} \frac{1}{(2r-1)^4} =$
 1) $\frac{16}{15}a$ 2) $\frac{a}{2}$ 3) $\frac{15}{16}a$ 4) $\frac{14}{15}a$
32. If $\sum T_r = \frac{n(n+1)(n+2)}{3}$ then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{2008}{T_r} =$
 1) 2008 2) 8002 3) 2080 4) 2000
33. The coefficient of x^{n-2} in
 $(x-1)(x-2)(x-3)\dots(x-n)$ is
 1) $\frac{n(n^2+1)(3n+2)}{24}$ 2) $\frac{n(n^2-1)(3n+2)}{24}$
 3) $n(n^2+1)(3n+2)$ 4) $n(n^2+1)$
34. The sum of 15 terms of the series
 $1 + (1+3) + (1+3+5) + (1+3+5+7) + \dots$ is
 $10 \times \lambda$ then value of $\lambda =$
 1) 124 2) 240 3) 220 4) 142

LEVEL-II (C.W) - KEY

- 1) 2 2) 3 3) 1 4) 1 5) 1 6) 4
 7) 4 8) 4 9) 2 10) 3 11) 4 12) 1
 13) 4 14) 4 15) 1 16) 3 17) 1 18) 1
 19) 2 20) 2 21) 2 22) 2 23) 1 24) 2
 25) 3 26) 1 27) 3 28) 3 29) 3 30) 1
 31) 3 32) 1 33) 2 34) 1

LEVEL-II (C.W) - HINTS

1. $a_1 + a_{2n} = a_2 + a_{2n-1} = a_n + a_{n+1} = K$ (say)
 $= K \left\{ \frac{\sqrt{a_1} - \sqrt{a_2}}{a_1 - a_2} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n+1}}}{a_n - a_{n+1}} \right\}$
 rationalising term $= \frac{-K}{d} \frac{a_1 - a_{n+1}}{\sqrt{a_1} + \sqrt{a_{n+1}}}$
 where $d = a_1 - a_2 = \dots = a_n - a_{n+1}$
2. $\frac{2n}{2} (4 + (2n-1)3) = \frac{n}{2} (114 + (n-1)2)$
3. $28, A_1, A_2, \dots, A_{11}, 10$ are in A.P
 $10 = T_{13} = 28 + 12d$

$$d = -\frac{3}{2} \quad \text{Integral A.M.'s are}$$

$$28 - 2\left(\frac{3}{2}\right), 28 - 4\left(\frac{3}{2}\right), 28 - 6\left(\frac{3}{2}\right),$$

$$28 - 8\left(\frac{3}{2}\right), 28 - 10\left(\frac{3}{2}\right)$$

No. of integral A.M.'s = 5

$$4. (2+0) + (2+1^2) + (2+2^2) + (2+3^2) + \dots$$

$$\therefore t_n = 2 + (n-1)^2$$

$$5. \text{ In A.P.'s } d_1 = 4 \text{ and } d_2 = 5$$

common term is 11, 31, 51, 71, ...

$$T_{10} = \{LCM \text{ of } (4, 5)\}(10-1) + 11 = 191$$

$$6. f(x) = px^2 + qx + r$$

$$f(1) = f(-1) \Rightarrow q = 0$$

$$\therefore f(x) = px^2 + r \Rightarrow f'(x) = 2px$$

$$7. \text{ Given that } ar^n = ar^{n+1} + ar^{n+2} \Rightarrow r^2 + r - 1 = 0$$

$$8. a - ar = 12 ; ar^2 - ar^3 = 48$$

$$9. \text{ Given quadratic expression } \geq 0, a > 0 \text{ and } x \in R \quad \therefore D \leq 0$$

$$\Rightarrow \left(\sum_{i=1}^3 a_i a_{i+1} \right) - \left(\sum_{i=1}^3 a_i^2 \right) \left(\sum_{i=2}^4 a_i^2 \right) \leq 0$$

$$\Rightarrow (a_1 a_2 + a_2 a_3 + a_3 a_4)^2$$

$$- (a_1^2 + a_2^2 + a_3^2)(a_2^2 + a_3^2 + a_4^2) \leq 0$$

$$\Rightarrow \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3}$$

$$10. f_n(x) = x.1 + x.10^2 + x.10^3 + \dots x.10^{n-1}$$

$$= \frac{x}{9}(10^n - 1) \quad \therefore f_n^2(3) + f_n(2)$$

$$= \left[\frac{3}{9}(10^n - 1) \right]^2 + \frac{2}{9}(10^n - 1) = \frac{10^{2n} - 1}{9} = f_{2n}(1)$$

$$11. 1 - \frac{1}{3} + 1 - \frac{1}{9} + 1 - \frac{1}{27} + \dots n \text{ terms}$$

$$= n - \left[\frac{\frac{1}{3} \left[1 - \left(\frac{1}{3} \right)^n \right]}{1 - \frac{1}{3}} \right]$$

$$12. x = \frac{1}{1-a}, y = \frac{1}{1-b}$$

$$13. \text{ Given that } \frac{a}{1-r} = 4 \text{ and } ar = \frac{3}{4}, \text{ solve}$$

$$14. \text{ Given that } \frac{a}{1-r} = 20, \frac{a^2}{1-r^2} = 100$$

$$15. 3(24) + 3(12) + 3(6) + \dots \infty$$

$$\Rightarrow 72 \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots \infty \right) = 144 \text{ cm}$$

$$16. \frac{1}{3} + \frac{1}{3^2} + \dots \infty = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{3} \times \frac{3}{2} = \frac{1}{2}$$

$$\text{and } y = (0.36)^{\log_4 \frac{1}{2}}$$

$$17. \text{ Let } \frac{A}{R}, A, AR \text{ be the roots of the equation}$$

$$ax^3 + bx^2 + cx + d = 0$$

$$\therefore \text{ Product of the roots } = -\frac{d}{a}$$

$$\Rightarrow A^3 = -\frac{d}{a} \Rightarrow A = -\left(\frac{d}{a}\right)^{1/3}$$

Since A is a root of the given equation.

$$\therefore aA^3 + bA^2 + cA + d = 0$$

$$\Rightarrow a \left(-\frac{d}{a} \right) + b \left(-\frac{d}{a} \right)^{2/3} + c \left(-\frac{d}{a} \right)^{1/3} + d = 0$$

$$\Rightarrow b \left(\frac{d}{a} \right)^{2/3} = c \left(\frac{d}{a} \right)^{1/3} \Rightarrow b^3 \frac{d^2}{a^2} = c^3 \frac{d}{a} \Rightarrow b^3 d = c^3 a$$

$$18. (a-c)^2 + (b-c)^2 + (b-d)^2 - (a-d)^2$$

$$= -2ac - 2bc - 2bd + 2ad + 2b^2 + 2c^2,$$

$$\left(\frac{a}{b} = \frac{b}{c} = \frac{c}{d} \right)$$

$$19. 2a = b + c \text{ and } b, G_1, G_2, c \text{ are in G.P.}$$

$$\text{Then } G_1 = br, G_2 = br^2, c = br^3 \text{ and } r = \left(\frac{c}{b} \right)^{1/3}$$

20. $\frac{x}{y}, \frac{y}{z}, \frac{z}{x}$ etc are positive $A.M \geq G.M$

21. If a,b,c are the sides of triangle then

$$b+c-a > 0, c+a-b > 0, a+b-c > 0$$

$$x = b+c-a, y = c+a-b, z = a+b-c$$

$$y+z=2a, z+x=2b, x+y=2c$$

Now. $\frac{y+z}{2x} + \frac{z+x}{2y} + \frac{x+y}{2z} \geq 3$

22. $\frac{2 \cdot \frac{3x}{2} + 3 \cdot \frac{4y}{3}}{5} \geq \left(\left(\frac{3x}{2} \right)^2 \left(\frac{4y}{3} \right)^3 \right)^{\frac{1}{5}} \Rightarrow x^2 y^3 \leq \frac{3}{16}$

23. By the cauchy- Schwartz's in equality

$$\left(\sqrt{4a+1} + \sqrt{4b+1} + \sqrt{4c+1} \right)^2 \leq$$

$$3[4(a+b+c)+3] \leq 81$$

24. $a = x + (p-1)d, b = x + (q-1)d,$

$$c = x + (r-1)d$$

$$a = mn^{p-1}, b = mn^{q-1}, c = mn^{r-1}$$

$\therefore$ Product of roots =

$$(mn^{p-1})^{(r-q)d} (mn^{q-1})^{(p-r)d} (mn^{r-1})^{(q-p)d}$$

$$= m^0 \cdot n^0 = 1$$

25. Suppose $p^a = q^b = r^c = s^d =$

K so that $p = K^{\frac{1}{a}}; q = K^{\frac{1}{b}}; r = K^{\frac{1}{c}}; s = K^{\frac{1}{d}}$

common ratio =

$$\frac{q}{p} = K^{\frac{1}{b} - \frac{1}{a}} = \frac{r}{q} = K^{\frac{1}{c} - \frac{1}{b}} = \frac{s}{r} = K^{\frac{1}{d} - \frac{1}{c}}$$

$$\therefore \frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b} = \frac{1}{d} - \frac{1}{c}, = a, b, c, d \text{ are in H.P.}$$

26. $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in AP

$$\therefore \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} = \dots = \frac{1}{a_n} - \frac{1}{a_{n-1}} = d$$

$$\Rightarrow \frac{a_1 - a_2}{a_1 a_2} = \frac{a_2 - a_3}{a_2 a_3} = \dots = \frac{a_{n-1} - a_n}{a_{n-1} a_n} =$$

$$\frac{a_1 - a_2 + a_2 - a_3 + \dots + a_{n-1} - a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n}$$

by law of proportion

27. $3(a^2 + b^2 + c^2 + 1) - 2(a + b + c)$

$$- 2(ab + bc + ca) = 0$$

$$\Rightarrow 2(a^2 + b^2 + c^2 - ab - bc - ca) +$$

$$(a^2 + b^2 + c^2 - 2a - 2b - 2c + 3) = 0$$

$$\Rightarrow [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$+ (a-1)^2 + (b-1)^2 + (c-1)^2 = 0$$

$$\Rightarrow a = b = c = 1$$

$\Rightarrow a, b, c$ are in A.P and G.P both

28. $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}, -\frac{y}{b} = \frac{x}{a} + \frac{1}{c}$

29. $\frac{2ab}{a+b} = 4, \frac{a+b}{2} = A$
 $\sqrt{ab} = G$

30. as n is odd the value of the given expression

$$1^3 - 2^3 + 3^3 - \dots + n^3$$

$$= 1^3 + 2^3 + \dots + n^3 - 2(2^3 + \dots + (n-1)^3)$$

31. $\sum_{r=1}^{\infty} \frac{1}{(2r-1)^4} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \text{to } \infty$

$$= a - \frac{1}{2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \text{to } \infty \right); a - \frac{a}{16} = \frac{15}{16} a$$

32. $\Rightarrow t_n = \sum_{r=1}^n t_r - \sum_{r=1}^{n-1} t_r = n(n+1)$

$$= 2008 \left[\sum_{r=1}^n \left(\frac{1}{r} - \frac{1}{r+1} \right) \right]$$

$$\therefore \lim_{n \rightarrow \infty} \sum \frac{2008}{t_n} = \lim_{n \rightarrow \infty} 2008 \left(1 - \frac{1}{n+1} \right) = 2008$$

$$33. \quad x^n + (-1)(1 + \dots + n)x^{n-1} + (-1)^2 \left(\sum 1.2 \right) x^{n-2} + \dots + (-1)^n n! x^0$$

for coefficient x^{n-2} is calculated as

$$(1 + 2 + \dots + n)^2 = (1^2 + 2^2 + \dots + n^2) +$$

2(sum of the product of numbers taken two at a times)

$$34. \quad t_n = n^2; \quad s_n = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$n = 15; \quad s_{15} = \frac{15 \times 16 \times 31}{6} = 1240 = 10\lambda$$

$$\lambda = 124$$

LEVEL-II (H.W)

1. Let the sequence $a_1, a_2, a_3, \dots, a_n$ form an A.P

$a_1^2 - a_2^2 + a_3^2 - a_4^2 + \dots + a_{2n-1}^2 - a_{2n}^2$ is equal to

$$1) \frac{n}{2n-1} (a_1^2 - a_{2n}^2) \quad 2) \frac{1}{2n-1} (a_1^2 - a_{2n}^2)$$

$$3) \frac{n}{n+1} (a_1^2 + a_{2n}^2) \quad 4) \frac{1}{2n+1} (a_1^2 - a_{2n}^2)$$

2. The sum to 101 terms of an A.P. is 1212. The middle term is

$$1) 6 \quad 2) 12 \quad 3) 24 \quad 4) 26$$

3. If $\log 2, \log(2^x - 1)$ and $\log(2^x + 3)$ are in AP, then the value of x is given by

$$1) \frac{5}{2} \quad 2) \log_2 5 \quad 3) \log_3 5 \quad 4) \log_5 3$$

4. If in AP, $a_7 = 9$ and if a_1, a_2, a_7 is least, then common difference is

$$1) \frac{11}{30} \quad 2) \frac{13}{10} \quad 3) \frac{32}{33} \quad 4) \frac{33}{20}$$

5. The number of common terms in two A.P's 2, 7, 12, 17, 500 terms and 1, 8, 15, 22, 300 terms is

$$1) 58 \quad 2) 60 \quad 3) 61 \quad 4) 63$$

6. In G.P. $(p+q)^{th}$ term is m , $(p-q)^{th}$ term is n , then p^{th} term is

$$1) nm \quad 2) \sqrt{nm} \quad 3) m/n \quad 4) \sqrt{m/n}$$

7. If a_1, a_2, a_3 are three positive consecutive terms of a GP with common ratio K . then all values of K for which the inequality

$a_3 > 4a_2 - 3a_1$, is satisfies

$$1) (1, 3) \quad 2) (-\infty, 1) \cup (3, \infty) \\ 3) (-\infty, \infty) \quad 4) (0, \infty)$$

8. The series

$$\frac{2x}{x+3} + \left(\frac{2x}{x+3} \right)^2 + \left(\frac{2x}{x+3} \right)^3 + \dots \text{to } \infty \quad \text{will}$$

have a definite sum when

$$1) -1 < x < 3 \quad 2) 0 < x < 1 \\ 3) x = 0 \quad 4) x > 3$$

9. If a, b, c, d, x are real and the roots of equation

$$(a^2 + b^2 + c^2)x^2 - 2(ab + bc + cd)x +$$

$$(b^2 + c^2 + d^2) = 0 \text{ real and equal, then}$$

a, b, c, d are in

$$1) \text{ A.P} \quad 2) \text{ G.P} \quad 3) \text{ H.P} \quad 4) \text{ None of these}$$

10. $(666\dots n \text{ digits})^2 + (888\dots n \text{ digits}) =$

$$1) \frac{4}{9}(10^n - 1) \quad 2) \frac{4}{9}(10^{2n} - 1)$$

$$3) \frac{4}{9}(10^n - 1)^2 \quad 4) \frac{4}{9}(10^n - 1)^2$$

11. Let $a = 111\dots 1$ (55 digits),

$$b = 1 + 10 + 10^2 + \dots + 10^4$$

$$c = 1 + 10^5 + 10^{10} + 10^{15} + \dots + 10^{50}, \text{ then}$$

$$1) a = b+c \quad 2) a = bc \quad 3) b = ac \quad 4) c = ab$$

12. The sum to infinity of $\frac{1}{7} + \frac{2}{7^2} + \frac{1}{7^3} + \frac{2}{7^4} + \dots$ is

$$1) 1/5 \quad 2) 7/24 \quad 3) 5/48 \quad 4) 3/16$$

13. If each term of an infinite G.P is twice the sum of the terms following it, then the common ratio of G.P is

$$1) \frac{1}{2} \quad 2) \frac{2}{3} \quad 3) \frac{1}{3} \quad 4) \frac{3}{2}$$

14. Sum of infinite No. of terms in G.P is 20 and sum of their squares is 100, then the common ratio of G.P is

$$1) \frac{1}{5} \quad 2) \frac{4}{5} \quad 3) \frac{2}{5} \quad 4) \frac{3}{5}$$

15. If 's' is the sum to infinite terms of a G.P. whose first term is 1, then the sum of n terms is

1) $s \left(1 - \left(1 - \frac{1}{s} \right)^n \right)$ 2) $\frac{1}{s} \left(1 - \left(1 - \frac{1}{s} \right)^n \right)$
 3) $1 - \left(1 - \frac{1}{s} \right)^n$ 4) $1 + \left(1 - \frac{1}{s} \right)^n$

16. If $r > 1$ and $x = a + a/r + a/r^2 + \dots$,

$$y = b + b/r + b/r^2 + \dots,$$

$$\text{And } z = c + c/r + c/r^2 + \dots,$$

Then value of xy/z^2 is

1) ab/c^2 2) abr/c 3) ab/c^2r 4) ab/c

17. If the A.M. and G.M. of two numbers are 13 and 12 respectively then the two numbers are

1) 8, 12 2) 8, 18 3) 10, 18 4) 12, 18

18. If $n!$, $3(n!)$ and $(n+1)!$ are in G.P., then $n!$, $5(n!)$ and $(n+1)!$ are in

1) A.P. 2) G.P. 3) H.P. 4) None

19. If G_1 and G_2 are two geometric means and A is the arithmetic mean inserted between two positive numbers then the value of

$$\frac{G_1^2}{G_2} + \frac{G_2^2}{G_1} \text{ is}$$

1) $A/2$ 2) A 3) $2A$ 4) $3A$

20. If $x_i > 0$, $i = 1, 2, 3, \dots, 50$ and

$$x_1 + x_2 + x_3 + \dots + x_{50} = 50 \text{ and minimum}$$

value of $\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_{50}}$ is λ then

$$\lambda =$$

1) 50 2) 60 3) 40 4) 202.

21. If $A_1, A_2, A_3, \dots$ belongs to A.P such that

$$A_1 + A_4 + A_7 + \dots + A_{28} = 140 \text{ then maximum}$$

value of $A_1 \cdot A_2 \cdot \dots \cdot A_{28}$ is

1) 2^{28} 2) 7^{28} 3) $(14)^{28}$ 4) $(28)^{28}$

22. Let a, b and c be the real numbers such that $a + b + c = 6$ then, the range of ab^2c^3 is

1) $(0, \infty)$ 2) $(0, 1)$
 3) $(0, 108]$ 4) $(6, 108]$

23. If none of $b_1, b_2, \dots, b_n$ is zero then

$$\left(\frac{a_1}{b_1} + \frac{a_2}{b_2} + \dots + \frac{a_n}{b_n} \right)^2 \text{ is}$$

1) $\geq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^{-2} + b_2^{-2} + \dots + b_n^{-2})$

2) $\leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^{-2} + b_2^{-2} + \dots + b_n^{-2})$

3) $> (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^{-2} + b_2^{-2} + \dots + b_n^{-2})$

4) $< (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^{-2} + b_2^{-2} + \dots + b_n^{-2})$

24. If a, b, c be the p^{th} , q^{th} and r^{th} terms respectively of a G.P., then the equation

$$a^q b^r c^p x^2 + pqr x + a^r b^p c^q = 0 \text{ has}$$

1) both roots zero

2) at least one root zero

3) no root zero

4) both roots unity

25. If $-1 < a, b, c < 1$ and a, b, c are in A.P.

and $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$ then x, y,

z are in

1) A.P. 2) G.P. 3) H.P. 4) A.G.P

26. If $a_1, a_2, a_3, \dots, a_n$ are in H.P then

$$\frac{a_1}{a_2 + a_3 + \dots + a_n}, \frac{a_2}{a_1 + a_3 + \dots + a_n}, \frac{a_3}{a_1 + a_2 + \dots + a_n},$$

$$\dots, \frac{a_n}{a_2 + a_3 + \dots + a_{n-1}}$$

1) A.P. 2) G.P. 3) H.P. 4) A.G.P

27. If a, 8, b are in A.P; a, 4, b are in G.P; a, x, b are in H.P then x =

1) 2 2) 1 3) 4 4) 16

28. Number of positive integral ordered pairs of (a, b) such that 6, a, b are in H.P is

1) 5 2) 6 3) 7 4) 8

29. If a, b, c are in H.P, then the value of $\frac{a+c}{a-c}$ is

1) $\frac{a}{a-b}$ 2) $\frac{a-b}{a}$ 3) $\frac{b}{a}$ 4) $\frac{a}{a+b}$

30. If $x > 1, y > 1, z > 1$ are in G.P then

$$\frac{1}{1 + \log x}, \frac{1}{1 + \log y}, \frac{1}{1 + \log z} \text{ are in}$$

1) AP 2) GP 3) HP 4) AGP

31. If $a = \sum_{r=1}^{\infty} \frac{1}{r^2}$, $b = \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}$, then $\frac{a}{b} =$

- 1) $\frac{5}{4}$ 2) $\frac{4}{3}$ 3) $\frac{3}{4}$ 4) $\frac{4}{5}$

LEVEL-II (H.W)-KEY

- 1) 1 2) 2 3) 2 4) 4 5) 2 6) 2
7) 2 8) 1 9) 2 10) 2 11) 2 12) 4
13) 3 14) 4 15) 1 16) 1 17) 2 18) 1
19) 3 20) 1 21) 3 22) 3 23) 2 24) 3
25) 3 26) 3 27) 1 28) 3 29) 1 30) 3 31) 2

LEVEL-II (H.W) - HINTS

- $-d(a_1 + a_2 + a_3 + \dots + a_{2n})$
- $S_{101} = 1212 \Rightarrow a + 50d = 12$, middle term $= \frac{T_{n+1}}{2}$
- t is given that
 $\log 2, \log(2^x - 1), \log(2^x + 3)$ are in A.P.
 $\Rightarrow 2\log(2^x - 1) = \log 2 + \log(2^x + 3)$
 $\Rightarrow (2^x - 1)^2 = 2(2^x + 3)$
 $\Rightarrow (2^x)^2 - 4(2^x) - 5 = 0$
 $\Rightarrow (2^x - 5)(2^x + 1) = 0 \Rightarrow 2^x = 5 \Rightarrow x = \log_2 5$
- $a_7 = 9 \Rightarrow a_1 + 6d = 9$; $D = a_1 a_2 a_7$
 $= (9 - 6d)(9 - 5d)9 = 270 \left[\left(d - \frac{33}{20} \right)^2 - \frac{9}{400} \right]$
is least for $d = \frac{33}{20}$
- 2, 7, 12, 17, 500 terms
 $T_{500} = 2 + (500 - 1)5 = 2497$
1, 8, 15, 22, 300 terms
 $T_{300} = 1 + (300 - 1)7 = 2094$
The common difference of common terms = $5 \times 7 = 35$
Common terms are 22, 57, 92,
Let last term ≤ 2094
 $\Rightarrow 22 + (n - 1)35 \leq 2094$
 $\Rightarrow n \leq 60.2$

6. $ar^{p+q-1} = m$ and $a.r^{p-q-1} = n$, find mn

7. $\frac{a_2}{a_1} = \frac{a_3}{a_2} = K$ From the given in equality

$$K^2 a_1 > 4a_1 K - 3a_1 > 0 \Rightarrow K^2 - 4K - 3 > 0$$

8. Common ratio of given $G.P. = \frac{2x}{x+3}$

For definite sum of infinite G.P., $-1 < \frac{2x}{x+3} < 1$

$$\Rightarrow \frac{2x}{x+3} + 1 > 0 \text{ and } \frac{2x}{x+3} - 1 < 0 \Rightarrow -1 < x < 3$$

9. Roots are real and equal

$$\Rightarrow (a^2 + b^2 + c^2)(b^2 + c^2 + d^2) - (ab + bc + cd)^2 = 0$$

$$\Rightarrow b^2 = ac, c^2 = bd, ac = bd$$

$$\Rightarrow a, b, c, d \text{ are in G.P.}$$

10. $(6 + 6(10) + \dots + 6(10)^{n-1})^2 + [8 + 8(10) + \dots + 8(10)^{n-1}]$

$$= \left(\frac{2}{3} (10^n - 1) \right)^2 + \frac{8}{9} (10^n - 1)$$

11. $a = 1 + 10 + 10^2 + \dots + 10^{54}$

$$\frac{10^{55} - 1}{10 - 1} = \frac{10^{55} - 1}{10^5 - 1} \cdot \frac{10^5 - 1}{10 - 1} = bc$$

12. $\left(\frac{1}{7} + \frac{1}{7^3} + \frac{1}{7^5} + \dots \right) + \left(\frac{2}{7^2} + \frac{2}{7^4} + \frac{2}{7^6} + \dots \right)$

$$S_{\infty} = a / 1 - r$$

13. $a_n = 2[a_{n+1} + a_{n+2} + a_{n+3} + \dots \infty], \forall n \in N$

$$ar^{n-1} = 2[ar^n + ar^{n+1} + ar^{n+2} + \dots];$$

$$ar^{n-1} = \frac{2ar^n}{1-r} \Rightarrow r = \frac{1}{3}$$

14. $a + ar + ar^2 + \dots \infty = 20 \Rightarrow \frac{a}{1-r} = 20 \dots (1)$

$$a^2 + a^2 r^2 + a^2 r^4 + \dots \infty = 100$$

$$\Rightarrow \frac{a^2}{1-r^2} = 100 \dots (2)$$

from 1 and 2 we get $r = \frac{3}{5}$

$$15. \quad s = \frac{1}{1-r} \Rightarrow r = \left(1 - \frac{1}{s}\right)$$

$$\text{sum to } n \text{ terms} = \frac{1 \left(1 - \left(1 - \frac{1}{s}\right)^n\right)}{1 - \left(1 - \frac{1}{s}\right)} = s \left(1 - \left(1 - \frac{1}{s}\right)^n\right)$$

$$16. \quad \text{we have } x = \frac{ar}{r-1}, y = \frac{br}{r-1}, z = \frac{cr}{r-1}$$

$$17. \quad \frac{a+b}{2} = 13$$

$$\sqrt{ab} = 12$$

$$18. \quad 9(n!)^2 = (n!)(n+1)! \Rightarrow n = 8$$

19. from synopsis

$$G_n = a \left(\frac{b}{a}\right)^{\frac{n}{n+1}}; \quad A_n = a + \frac{n(b-a)}{n+1}$$

$$20. \quad \frac{x_1 + x_2 + \dots + x_{50}}{50} \geq (x_1 x_2 \dots x_n)^{\frac{1}{50}} \dots (1)$$

$$\frac{\frac{1}{x_1} + \dots + \frac{1}{x_{50}}}{50} \geq \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \dots \frac{1}{x_{50}}\right)^{\frac{1}{50}} \dots (2)$$

$$\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}} \geq 50 \therefore \lambda = 50$$

$$21. \quad A_1 + A_4 + A_7 + \dots + A_{28} = 140$$

$$A_1 + A_{28} = A_4 + A_{25} = \dots = A_{13} + A_{16}$$

$$5(A_1 + A_{28}) = 140 \Rightarrow A_1 + A_{28} = 28$$

$$\frac{A_1 + A_2 + \dots + A_{28}}{28} = 14$$

$$AM \geq GM$$

$$22. \quad \frac{a + 2\left(\frac{b}{2}\right) + 3\left(\frac{c}{3}\right)}{6} \geq \left(a \left(\frac{b}{2}\right)^2 \left(\frac{c}{3}\right)^3\right)^{\frac{1}{6}}$$

$$1 > \left(\frac{ab^2c^3}{108}\right)^{\frac{1}{6}} \Rightarrow ab^2c^3 \leq 108$$

23. By using Cauchy-Schwartz's Inequality

$$\left(a_1 \cdot \frac{1}{b_1} + a_2 \cdot \frac{1}{b_2} + \dots + a_n \cdot \frac{1}{b_n}\right)^2$$

$$\leq (a_1^2 + a_2^2 + \dots + a_n^2) \left(\frac{1}{b_1^2} + \frac{1}{b_2^2} + \dots + \frac{1}{b_n^2}\right)$$

24. Product of roots = $a^{r-q} b^{p-r} c^{q-p} = 1 \neq 0$
no root is equal to zero.

$$25. \quad x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c}$$

find a, b, c

given a, b, c are in A.P.

$$26. \quad \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3} \dots \dots \dots \text{ in AP}$$

$$\frac{a_1 + a_2 + a_3 + \dots + a_n}{a_1}, \frac{a_1 + a_2 + a_3 + \dots + a_n}{a_2}, \dots \dots \dots \frac{a_1 + a_2 + a_3 + \dots + a_n}{a_n}$$

are in AP

$$27. \quad a + b = 16 \text{ and } ab = 16 \text{ and } \frac{2}{x} = \frac{1}{a} + \frac{1}{b}$$

$$28. \quad 6, a, b \text{ are in H.P} \Rightarrow \frac{1}{6}, \frac{1}{a}, \frac{1}{b} \text{ are in A.P}$$

$$\Rightarrow \frac{2}{a} = \frac{1}{6} + \frac{1}{b} \Rightarrow b = \frac{6a}{12-a}$$

$$a \in \{3, 4, 6, 8, 9, 10, 11\}$$

$$29. \quad \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in A.P. } \frac{2}{b} = \frac{1}{c} + \frac{1}{a} = \frac{a+c}{ac}$$

$$\frac{a+c}{a-c} = \frac{2ac}{b(a-c)} = \frac{\frac{2}{b}}{\frac{1}{c} - \frac{1}{a}}$$

$$30. \quad y^2 = zx \Rightarrow 1 + \log x, 1 + \log y, 1 + \log z \text{ are in AP}$$

$$31. \quad a = \frac{1}{1^2} + \frac{1}{3^2} + \dots + \frac{1}{4} \left(\frac{1}{1^2} + \frac{1}{2^2} + \dots\right)$$

$$a - \frac{a}{4} = b \Rightarrow \frac{a}{b} = \frac{4}{3}$$

LEVEL - III

1. The series of natural numbers are arranged

as follow.

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 2 & & 3 & \\
 & 4 & & 5 & & 6 & \\
 7 & & 8 & & 9 & & 10
 \end{array}$$

sum of numbers in the n th row is

- 1) $\frac{n(n+1)}{2}$ 2) $\frac{n(n^2+1)}{2}$ 3) $\frac{n^2(n+1)}{2}$ 4) $\frac{n^2(n^2+1)}{2}$
2. If a, b, c, d are distinct integers in A.P. such that $d = a^2 + b^2 + c^2$, then $a + b + c + d =$
 1) 0 2) 1 3) 2 4) 4
3. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute.
 If $a_1 = a_2 = \dots = a_{10} = 150$ and $a_{10}, a_{11}, \dots$ are in A.P. with common difference -2, then the time taken by him to count all notes is
 1) 135mins 2) 24mins 3) 34mins 4) 125mins
4. A man saves Rs. 200 in each of the first 3 months of his service. In each of the subsequent months his saving increases by Rs. 40 more than the saving of immediately previous month. His total saving from the start of service will be Rs 11040 after.
 [AIEEE 2011]
 1) 21 months 2) 18 months
 3) 19 months 4) 20 months
5. The sum of first 20 terms of the sequence 0.7, 0.77, 0.777, is. [MAINS-2013]
 1) $\frac{7}{81}(179 - 10^{-20})$ 2) $\frac{7}{9}(99 - 10^{-20})$
 3) $\frac{7}{81}(179 + 10^{-20})$ 4) $\frac{7}{9}(99 + 10^{-20})$
6. Sum of n terms of the series 1, 3, 7, 15, 31, is
 1) $2^{n+1} - n - 2$ 2) $2^n - n - 2$
 3) $2^{n+1} + n + 2$ 4) $2^n - 1$

7. The three successive terms of a GP will form the sides of a triangle if the common ratio satisfies the inequality ($r > 1$)

1) $\left(1, \frac{\sqrt{5}+1}{2}\right)$ 2) $\left(-\infty, \frac{\sqrt{5}-1}{2}\right) \cup \left(\frac{\sqrt{5}+1}{2}, \infty\right)$
 3) $[-\sqrt{5}, \sqrt{5}]$ 4) $(-\sqrt{5}, \sqrt{5})$

8. If a, b, c be respectively the $p^{\text{th}}, q^{\text{th}}$ and r^{th}

terms of G.P then $\Delta = \begin{vmatrix} \log a & \log b & \log c \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$

equals to

1) 1 2) 0 3) -1 4) 2

9. If $t_r = 2^{\frac{r}{3}} + 2^{-\frac{r}{3}}$, then $\sum_{r=1}^{100} t_r^3 - 3 \sum_{r=1}^{100} t_r + 1 =$

1) $\frac{2^{101}+1}{2^{100}}$ 2) $\frac{2^{101}-1}{2^{100}}$ 3) $\frac{2^{201}-1}{2^{100}}$ 4) $\frac{2^{201}+1}{2^{100}}$

10. The value of x satisfying the equation

$$\left[3 \left(1 - \frac{1}{2} + \frac{1}{4} \dots \text{to } \infty \right) \right]^{\log_{10} x} = \left[20 \left(1 - \frac{1}{4} + \frac{1}{16} \dots \text{to } \infty \right) \right]^{\log_x 10} \text{ is }$$

1) $\frac{1}{100}$ 2) 10 3) 1000 4) $\frac{1}{10}$

11. If $\exp \{ (\sin^2 x + \sin^4 x + \sin^6 x + \dots \text{upto } \infty) \log_e 2 \}$ satisfies the equation $x^2 - 17x + 16 = 0$

then the value of $\frac{2 \cos x}{\sin x + 2 \cos x}$ ($0 < x < \pi/2$) is

1) 1/2 2) 3/2 3) 5 4) 2/3

12. The length of the side of square is ' a ' metre. A second square is formed by joining the middle points of the sides of the squares. Then a third square is formed by joining the middle points of the sides of the second squares and so on. Then the sum of the area of squares which carried upto infinity is

1) a^2 2) $2a^2$ 3) $3a^2$ 4) $4a^2$

13. If $\frac{a+be^y}{a-be^y} = \frac{b+ce^y}{b-ce^y} = \frac{c+de^y}{c-de^y}$ then a,b,c,d are in
1) A.P. 2) G.P. 3) H.P. 4) A.G.P
14. If a,b,c,d are positive real numbers such that $a+b+c+d=2$, then $M=(a+b)(c+d)$ satisfies the relation
1) $0 < M \leq 1$ 2) $1 \leq M \leq 2$
3) $2 \leq M \leq 3$ 4) $3 \leq M \leq 4$
15. If n be the number of sequence a,b,c,d,e satisfying the conditions
(i) a,b,c,d,e are in A.P and G.P. both,
(ii) $c=3, 7$ then 'n' = -----
1) 1 2) 2 3) 5 4) 10
16. If $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms of an A.P are in G.P. whose common ratio is k, then the root of equation $(q-r)x^2 + (r-p)x + (p-q) = 0$ other than unity is
1) k 2) 2k 3) k^2 4) $\frac{1}{k}$
17. If the sum to infinity of the series $1+4x+7x^2+10x^3+\dots$ is $\frac{35}{16}$ then x =
1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{7}$ 4) $\frac{1}{7}$
18. The value of $2^{1/4} 4^{1/8} 8^{1/16} 16^{1/32} \dots$ is
1) 2 2) $3/2$ 3) 1 4) $1/2$
19. Let x be the arithmetic mean and y,z be the two geometric means between any two positive numbers. Then value of $\frac{y^3+z^3}{xyz}$ is
1) 2 2) 3 3) $1/2$ 4) $3/2$
20. If a, b, c are in G.P., then the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root if a/d, b/e, c/f are in
1) A.P. 2) G.P. 3) H.P. 4) A.G.P
21. Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$. Then $I_2 + I_4, I_3 + I_5, I_4 + I_6, I_5 + I_7, \dots$ are in
1) A.P. 2) G.P. 3) H.P. 4) A.G.P
22. Let $a_1, a_2, \dots, a_{10}$ be in A.P. and $h_1, h_2, \dots, h_{10}$ be in H.P. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, then $a_4 h_7$ is
1) 2 2) 3 3) 5 4) 6
23. If the system of linear equations $x + 2ay + az = 0, x + 3by + bz = 0, x + 4cy + cz = 0$ has a non-zero solution, then a, b, c are in
1) G.P. 2) H.P.
3) Satisfy $a + 2b + 3c = 0$ 4) A.P.
24. If $\cos(x-y), \cos x$ and $\cos(x+y)$ are in H.P., then value of $\cos x \sec(y/2)$ is
1) $\pm\sqrt{2}$ 2) $\pm\sqrt{3}$ 3) ± 2 4) ± 1
25. If a, b, c are real and in A.P. and a^2, b^2, c^2 are in H.P., then
1) $a = b = c$ 2) $2b = 3a + c$
3) $b^2 = \sqrt{ac}/8$ 4) $ab = c$
26. If 9 A.M.'s and 9 H.M's be inserted between 2 and 3 and A be any A.M. and H be the corresponding H.M., then $H(5-A) =$
1) 10 2) 6 3) -6 4) -10
27. Suppose 'a' is a fixed real number such that $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$ if p,q, r are in AP then x,y,z all are in
1) A.P. 2) G.P. 3) H.P. 4) A.G.P.
28. a,b,c are in A.P; b,c,d are in G.P and c,d,e are in H.P. If $a=2$ and $e=18$, then the sum of all possible values of c is
1) -6 2) 6 3) 12 4) 0
29. If an A.P., a G.P. and a H.P. have the same first term and same $(2n+1)^{\text{th}}$ term and their $(n+1)^{\text{th}}$ terms are a,b,c, respectively, then the radius of the circle.
 $x^2 + y^2 + 2bx + 2ky + ac = 0$ is
1) k 2) $|k|$ 3) $\sqrt{b^2 - ac}$ 4) k^2
30. If $a, a_1, a_2, a_3, a_4, \dots, a_{2n}, b$ are in A.P and $a, g_1, g_2, g_3, g_4, \dots, g_{2n}, b$ are in G.P and h is the H.M of a and b then $\frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + \dots + \frac{a_n + a_{n+1}}{g_n g_{n+1}}$ is equal to
1) $2n/h$ 2) $2nh$ 3) nh 4) n/h

31. If $f(x) = x^2 - (a+b)x + ab$ and A and H be the A.M. and H.M. between two quantities a and b, then

- 1) $Af(A) = Hf(H)$
- 2) $Af(H) = Hf(A)$
- 3) $A + f(A) = H + f(H)$
- 4) $f(A) + H = f(H) + A$

32. If positive numbers a, b, c be in H.P., then equation

$$x^2 - kx + 2b^{101} - a^{101} - c^{101} = 0 \quad (k \in \mathbb{R}) \text{ has}$$

- 1) both roots positive
- 2) both roots negative
- 3) one positive & one negative root
- 4) both roots imaginary

33. The value of $\sum_{n=1}^{10} \int_0^n x dx$ is

- 1) an even integer
- 2) an irrational number
- 3) a rational number
- 4) an irrational number

34. Let $\sum_{r=1}^n r^4 = f(n)$, then $\sum_{r=1}^n (2r-1)^4$ is equal to

- 1) $f(2n) - 16f(n)$
- 2) $f(2n) - 7f(n)$
- 3) $f(2n-a) - 8f(n)$
- 4) $f(2n-a) - 7f(n)$

35. For $x \in \mathbb{R}$ let $[x]$ denote the greatest integer $\leq x$. Largest natural number n for which

$$E = \left[\frac{\pi}{2} \right] + \left[\frac{1}{100} + \frac{\pi}{2} \right] + \left[\frac{2}{100} + \frac{\pi}{2} \right] + \dots + \left[\frac{n}{100} + \frac{\pi}{2} \right] < 43, \text{ is}$$

- 1) 41
- 2) 42
- 3) 43
- 4) 97

36. The sum to n terms of the series

$$\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots \text{ is}$$

- 1) $\frac{6n}{n+1}$
- 2) $\frac{9n}{n+1}$
- 3) $\frac{12n}{n+1}$
- 4) $\frac{3n}{n+1}$

37. Let r^{th} term of a series be given by

$$T_r = \frac{r}{1 - 3r^2 + r^4} \text{ then } \lim_{n \rightarrow \infty} \sum_{r=1}^n T_r =$$

- 1) $\frac{3}{2}$
- 2) $\frac{1}{2}$
- 3) $\frac{-1}{2}$
- 4) $\frac{-3}{2}$

38. The sum of the first n terms of the series

$$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots \text{ is } \frac{n(n+1)^2}{2}$$

when n is even. When n is odd the sum is

- 1) $\frac{3n(n+1)}{2}$
- 2) $\left[\frac{n(n+1)}{2} \right]^2$
- 3) $\frac{n(n+1)^2}{4}$
- 4) $\frac{n^2(n+1)}{2}$

39. Sum to n terms of the series

$$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \dots \text{ is}$$

- 1) $\tan^{-1}\left(\frac{n}{n+2}\right)$
- 2) $\tan^{-1}\left(\frac{2n-1}{2n+2}\right)$
- 3) $\tan^{-1}\left(\frac{1}{3n}\right)$
- 4) $\tan^{-1}\left(\frac{n}{n+1}\right)$

40. The sum of the series

$$\frac{1}{3^2 + 1} + \frac{1}{4^2 + 2} + \frac{1}{5^2 + 3} + \frac{1}{6^2 + 4} + \dots \infty \text{ is}$$

- 1) $\frac{13}{36}$
- 2) $\frac{13}{33}$
- 3) $\frac{11}{36}$
- 4) $\frac{15}{36}$

41. $\sum_{n=1}^n n(1-a)(1-2a)(1-3a) \dots \{1-(n-1)a\} =$

- 1) $1 - (1-a)(1-2a)(1-3a) \dots (1-na)$
- 2) $a[1 - (1-a)(1-2a) \dots (1-na)]$
- 3) $\frac{1}{a}[1 - (1-a)(1-2a) \dots (1-na)]$
- 4) $\frac{1}{a}[1 - (1-a)(1-2a)(1-3a) \dots \{1-(n-1)a\}]$

42. If $\sum_{r=1}^n t_r = \sum_{k=1}^n \sum_{j=1}^k \sum_{i=1}^j 2$, then $\sum_{r=1}^n \frac{1}{t_r} =$

- 1) $\frac{n+1}{n}$
- 2) $\frac{n}{n+1}$
- 3) $\frac{n-1}{n}$
- 4) $\frac{n}{n-1}$

43. $S_n = \sum_{n=1}^n \frac{n}{1+n^2+n^4}$, then $S_{10} \cdot S_{20}$

- 1) $\frac{110}{111} \cdot \frac{211}{421}$
- 2) $\frac{110}{421} \cdot \frac{111}{112}$
- 3) $\frac{110}{111} \cdot \frac{420}{421}$
- 4) $\frac{55}{111} \cdot \frac{210}{421}$

Since $n=125$ is not possible, total time required is $24+10=34$ minutes.

4. Total saving = $200+200+200+240+280+\dots$ to n months = 11040

$$\Rightarrow 400 + \frac{n-2}{2} [400 + (n-3)40] = 11040$$

$$\Rightarrow (n-21)(n+26) = 0 \Rightarrow n = 21$$

5. $0.7 + 0.77 + 0.777 + \dots + 0.777\dots 7$

$$\Rightarrow \frac{7}{9} [0.9 + 0.99 + \dots + 0.999\dots 9]$$

$$\Rightarrow \frac{7}{9} [(1-0.1) + (1-0.01) + \dots + (1-0.000\dots 1)]$$

$$\Rightarrow \frac{7}{9} \left[20 - \left(\frac{1}{10} + \frac{1}{10^2} + \dots + \frac{1}{10^{20}} \right) \right]$$

$$= \frac{7}{81} (179 + 10^{-20})$$

6. Let $S = 1 + 3 + 7 + 15 + 31 + \dots + T_n$ (1)

$$S = 0 + 1 + 3 + 7 + 15 + \dots + T_{n-1} + T_n$$
(2)

$$(1)-(2) \Rightarrow T_n = \frac{1(2^n-1)}{(2-1)} = 2^n - 1 \text{ and } S_n = \sum T_n$$

7. Sum of two sides of a triangle > third side

$$a + ar > ar^2$$

8. Let A be the first term and R be the common ratio of the G.P. Then,

$$a = AR^{p-1} \Rightarrow \log a = \log A + (p-1) \log R \dots (i)$$

$$b = AR^{q-1} \Rightarrow \log b = \log A + (q-1) \log R \dots (ii)$$

$$c = AR^{r-1} \Rightarrow \log c = \log A + (r-1) \log R \dots (iii)$$

Multiplying (i), (ii) and (iii) by $(q-r)$, $(r-p)$

and $(p-q)$ respectively and adding, we get

$$(q-r) \log a + (r-p) \log b + (p-q) \log c = 0$$

$$\Rightarrow \Delta = 0$$

9. $\sum_{r=1}^{100} t_r^3 = \sum_{r=1}^{100} 2^r + \sum_{r=1}^{100} \frac{1}{2^r} + 3 \sum_{r=1}^{100} t_r$

$$= 2^{101} - 2 + 1 - \frac{1}{2^{100}} + 3 \sum_{r=1}^{100} t_r$$

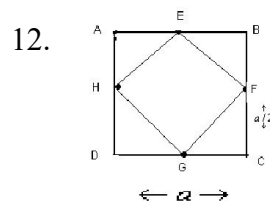
$$= \frac{2^{201} - 1}{2^{100}} - 1 + 3 \sum_{r=1}^{100} t_r$$

$$10. \left(3 \frac{1}{1+\frac{1}{2}} \right)^{\log_{10} x} = \left(\frac{20}{1+\frac{1}{4}} \right)^{\log_x^{10}}$$

$$2^{\log_{10} x} = (2^4)^{\log_x^{10}} \Rightarrow \log_{10} x = \frac{4}{\log_{10} x}$$

$$\therefore \log_{10} x = \pm 2 \Rightarrow x = 100 \text{ or } x = \frac{1}{100}$$

$$11. e^{\frac{\sin^2 x}{\cos^2 x} \times \log_e 2} = 16 \text{ or } 1 ; 2^{\tan^2 x} = 2^4 \text{ or } 2^0$$



- 12.

side of second square is $\frac{a}{\sqrt{2}}$, side of third square

is $\frac{a}{2}$, ... sum of areas of squares

$$a^2 + \left(\frac{a}{\sqrt{2}} \right)^2 + \left(\frac{a}{2} \right)^2 + \dots = 2a^2$$

$$13. \frac{2a - (a - be^y)}{a - be^y} = \frac{2b - (b - ce^y)}{b - ce^y}$$

$$= \frac{2c - (c - de^y)}{c - de^y} \text{ by law of proportion}$$

$$\Rightarrow \frac{(a - be^y)}{a} = \frac{(b - ce^y)}{b} = \frac{(c - de^y)}{c}$$

$$\Rightarrow a, b, c, d, \text{ are in G.P.}$$

14. Since G.M. $\leq$ A.M.

$$\therefore \sqrt{(a+b)(c+d)} \leq \frac{(a+b) + (c+d)}{2} = \frac{2}{2} = 1$$

Also $a, b, c, d > 0 \therefore M > 0$ Thus $0 < M \leq 1$.

15. a, b, c, d, e are in A.P. and G.P. both
 $\Rightarrow a = b = c = d = e = 3, 7$
 $\Rightarrow$ Required sequences are 3, 3, 3, 3, 3 and 7, 7, 7, 7, 7 $\Rightarrow n = 2$
16. Given $k = \frac{a+(q-1)d}{a+(p-1)d} = \frac{a+(r-1)d}{a+(q-1)d}$

$$= \frac{a+(q-1)d - a - (r-1)d}{a+(p-1)d - a - (q-1)d}$$

$$= \frac{(q-r)d}{(p-q)d} = \frac{(q-r)}{p-q} = \frac{1}{k}$$
17. $S_n = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$
18. $2^{\frac{1}{4}} \cdot 2^{\frac{1}{4}} \cdot 2^{\frac{3}{16}} \cdot 2^{\frac{1}{8}} \dots = 2^{\frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \frac{4}{2^5} + \dots}$
19. Given that $x = \frac{a+b}{2}$ and a, y, z, b are in G.P.
 $y^2 = az, z^2 = by, y^3 + z^3 = \frac{y^2}{xz} + \frac{z^2}{xy}$
20. $ax^2 + 2bx + c = 0 \Rightarrow (\sqrt{a}x + \sqrt{c})^2 = 0$
 $x = -\sqrt{\frac{c}{a}},$ use in $dx^2 + 2ex + f = 0$
21. We know that $I_n + I_{n+2} = \frac{1}{n+1}$ from integration
22. let 'd' is common difference of A.P.
 $\therefore 3 = a_{10} = 2 + 9d \Rightarrow d = \frac{1}{9}$
 let 'D' is common difference of $\frac{1}{h_1}, \frac{1}{h_2}, \dots, \frac{1}{h_{10}}$
 $\therefore \frac{1}{3} = \frac{1}{h_{10}} = \frac{1}{2} + \frac{9}{D} \Rightarrow D = \frac{-1}{54}$
23. $\det = 0$
24. $\frac{2}{\cos x} = \frac{1}{\cos(x+y)} + \frac{1}{\cos(x-y)}$
25. $2b = a + c$ & $b^2 = \frac{2a^2c^2}{a^2+c^2}$
 simplify, we get $a = 3 \therefore a = b = c$

26. Let A be the k^{th} A.M., then H will be the k^{th} H.M Now, $A = 2 + kd = 2 + k\left(\frac{3-2}{10}\right) = \frac{20+k}{10}$

$$H = \frac{1}{\frac{1}{2} + \frac{k\left(\frac{1}{3} - \frac{1}{2}\right)}{10}} = \frac{30-k}{60}$$

 $\therefore A + \frac{6}{H} = 5 \Rightarrow H(5-A) = 6$
27. $p-q = q-r = k$ (let) $\Rightarrow \frac{\frac{a}{x}-1}{p} = \frac{\frac{a}{y}-1}{q} = \frac{\frac{a}{z}-1}{r}$

$$\Rightarrow \frac{\left(\frac{a}{x}-1\right) - \left(\frac{a}{y}-1\right)}{p-q} = \frac{\left(\frac{a}{y}-1\right) - \left(\frac{a}{z}-1\right)}{q-r}$$

 by law of proportion $\Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{y} - \frac{1}{z}$
28. $b = \frac{a+c}{2}, c^2 = bd, d = \frac{2ce}{c+e}$
 Now, $c^2 = bd \Rightarrow c^2 = \left(\frac{a+c}{2}\right)\left(\frac{2ce}{c+e}\right)$
 $\Rightarrow c^2 = ae = 36; c = 6 \text{ or } -6$
29. let A be the first term, D be the common difference and B be the $(2n+1)^{\text{th}}$ term of A.P.
 then $B = A + 2nD \Rightarrow D = \frac{B-A}{2n}$
 $a = A + (n+1-1)D = \frac{A+B}{2}$
 similarly $b = \sqrt{AB}$ and $c = \frac{2AB}{A+B}$
 $\therefore b^2 = ac$ then find r.
30. $a+b = a_1 + a_{2n} = a_2 + a_{2n-1} = \dots$ and
 $ab = g_1 g_{2n} = g_2 g_{2n-1} \dots$ and $h = \frac{2ab}{a+b}$
31. We have to calculate $\frac{f(A)}{f(H)}$ and $f(A) - f(H)$
 Here $A = \frac{a+b}{2}, H = \frac{2ab}{a+b}; \frac{(a+b)^2}{4ab} = \frac{A}{H}$

32. a, b, c are in H.P $\Rightarrow$ H.M. of a and c is b
 $\Rightarrow \sqrt{ac} > b [\because G.M > H.M]$

Since A.M $>$ G.M. $a^{101} + c^{101} > 2(\sqrt{ac})^{101} > 2b^{101}$

$$f(x) = x^2 - kx + 2b^{101} - a^{101} - c^{101}$$

$$\text{Then } f(-\infty) > \infty > 0,$$

$$f(0) = 2b^{101} - a^{101} - c^{101} < 0$$

Hence equation $f(x) = 0$ has one root in $(-\infty, 0)$ and other in $(0, \infty)$

$$33. \sum_{n=1}^{10} \left(\frac{x^2}{2} \right)^n = \frac{1}{2} \sum_{n=1}^{10} n^2 = \frac{n(n+1)(2n+1)}{12}$$

is rational number

$$34. \sum_{r=1}^n (2r-1)^4 = \text{Total sum} - \text{Even sum} =$$

$$\sum_{r=1}^{2n} r^4 - \sum_{r=1}^n (2r)^4 = f(2n) - 16f(n)$$

$$35. \text{ Since } 3.14 < \pi < 3.142, 1.57 < \frac{\pi}{2} < 1.571$$

$$\therefore \left[\frac{\pi}{2} + \frac{n}{100} \right] = 1 \text{ for } n = 0, 1, 2, \dots, 42$$

the largest possible number n for which $E < 43$ is 41.

$$36. T_n = \frac{(2n+1)6}{n(n+1)(2n+1)} = 6 \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$S_n = \sum T_n = \frac{6n}{n+1}$$

$$37. T_r = \frac{r}{(r-1)^2 - r^2} = \frac{1}{2} \left[\frac{1}{r^2 - r - 1} - \frac{1}{r^2 - r + 1} \right]$$

$$38. \text{ If } n \text{ is odd; } S_{2m+1} = \frac{2m(2m+1)^2}{2} + (2m+1)^2$$

$$39. \tan^{-1} \left(\frac{2-1}{1+1.2} \right) + \tan^{-1} \left(\frac{3-2}{1+2.3} \right) + \dots \text{ Verify}$$

$$40. T_n = \frac{1}{n^2 + (n-2)} = \frac{1}{(n+2)(n-1)}, \text{ where}$$

$$n = 3, 4, 5, \dots \therefore S_{\infty} = \sum_{n=3}^{\infty} \frac{1}{3} \left[\frac{1}{n-1} - \frac{1}{n+2} \right] = \frac{13}{36}$$

$$41. t_n = \frac{1}{a} \left[\frac{(1-a)(1-2a)\dots\{1-(n-1)a\}}{-(1-a)(1-2a)\dots(1-na)} \right]$$

Put $n=1, 2, 3, \dots, n$ and add.

$$42. \sum_{k=1}^n \sum_{j=1}^k \sum_{i=1}^j 2 = \sum_{k=1}^n \sum_{j=1}^n 2j$$

$$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

$$\therefore S_n = \frac{n(n+1)(2n+1)}{3}$$

$$\Rightarrow t_r = S_r - S_{r-1} = \frac{r(r+1)(r+2)}{3} - \frac{(r-1)(r+1)}{3}$$

$$\therefore \sum_{r=1}^n \frac{1}{t_r} = 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

$$43. t_n = \frac{n}{(1+n^2)^2 - n^2} \Rightarrow S_n = \frac{1}{2} \left[1 - \frac{1}{1+n+n^2} \right]$$

$$\text{and } \Rightarrow S_{20} = \frac{1}{2} \left[1 - \frac{1}{421} \right] = \frac{210}{421}$$

$$44. \sum a_i b_i = \sum a_i (1 - a_i) = na - \sum a_i^2$$

$$= na - \sum (a_i - a)^2 - \sum a^2 - 2a \sum (a_i - a)$$

$$\Rightarrow \sum a_i b_i + \sum (a_i - a)^2 = nab$$

$$(\because \sum b_i = \sum 1 - \sum a_i, \therefore nb = n - na \text{ (or) } a+b=1)$$

$$45. \text{ Sum of first } n \text{ odd natural numbers} = n^2$$

$$\therefore \left(\frac{p+1}{2} \right)^2 + \left(\frac{q+1}{2} \right)^2 = \left(\frac{r+1}{2} \right)^2$$

$$\therefore p+1=8, q+1=6, r+1=10$$

$$46. T_n = \frac{n^2}{500 + 3n^3};$$

$$\frac{dT_n}{dn} = \frac{n(1000 - 3n^3)}{(500 + 3n^3)^2} = 0$$

$$n = \left(\frac{1000}{3} \right)^{\frac{1}{3}} \text{ between } 6 \text{ and } 7$$

Hence T_7 is largest term

47. Let the n consecutive odd integers be $2k+1, 2k+3, 2k+5, \dots, 2k+2n-1$

$$\text{Given } (n+k)^2 - k^2 = 25^2 - 11^2$$

$$\therefore k = 11, n+k = 25 \Rightarrow n = 14$$

48. $a_{n+2} + a_n - 2a_{n+1} = 0$

$$a_2 + a_{102} = 2a_{52}, a_3 + a_{103} = 2a_{53}.$$

49. In the given Sequence 1st term is 1.

The first 2 is in term 2

The first 4 is in term 4

The first 8 is in term 8

The sequence is doubling the first number and putting that number in the sequence for however many terms it is worth, i.e 8 is in the sequence 8 times, 4 is in the sequence 4 times, because we double the number each time, we know the pattern will go 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024,

So that means the number 1024 will start from 1024th term

$$\therefore 1025 \text{ term is also } 1024 = 2^{10}$$

50. $\tan^{-1} y - \tan^{-1} x = \tan^{-1} z - \tan^{-1} y$

$$\frac{y-x}{1+xy} = \frac{z-y}{1+zy} \quad \dots\dots\dots (1)$$

x, y, z are in A.P

$$y - x = z - y \quad \dots\dots\dots (2)$$

$$\text{from (1) and (2) } 1+xy = 1+zy$$

$$x=z \quad \therefore x=y=z \quad \therefore x, y, z \text{ are in A.P}$$



LEVEL-IV

In this section each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each of these questions has following four choices (1), (2), (3) and (4), only one of which is the correct answer.

1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

2) Statement -1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.

3) Statement-1 is True, Statement-2 is false

4) Statement-1 is False, Statement-2 is True.

1. **Statement-1:** If $a_1, a_2, a_3, \dots, a_n, \dots$ is an A.P. such that $a_1 + a_4 + a_7 + \dots + a_{16} = 147$, then

$$a_1 + a_6 + a_{11} + a_{16} = 98$$

Statement-2: In an A.P., the sum of the terms equidistant from the beginning and the end is always same and is equal to the sum of first and last term.

- 1) 1 2) 2 3) 3 4) 4

2. **Let** $a_1, a_2, a_3, \dots, a_{n-1}, a_n$ **be an A.P.**

$$\text{Statement-1: } a_1 + a_2 + a_3 + \dots + a_n = \frac{n}{2}(a_1 + a_n)$$

$$\text{Statement-2: } a_k + a_{n-k+1} = a_1 + a_n \text{ for}$$

$$k = 1, 2, 3, \dots, n$$

- 1) 1 2) 2 3) 3 4) 4

3. **Statement-1:** There exists no A.P. whose three terms are $\sqrt{3}, \sqrt{5}$ and $\sqrt{7}$.

Statement-2: If a_p, a_q and a_r are three distinct terms of an A.P., then $\frac{a_p - a_q}{a_p - a_r}$ is a rational number.

- 1) 1 2) 2 3) 3 4) 4

4. **Statement-1:** If for any real $x, 2^{1+x} + 2^{1-x}, \lambda$ and $3^x + 3^{-x}$ are three equidistant terms of an A.P., then $\lambda \geq 3$

Statement-2: A.M $\geq$ G.M for

- 1) 1 2) 2 3) 3 4) 4

5. **Statement-1:** If $x > 1$, the sum to infinite

$$\text{series } 1 + 3\left(1 - \frac{1}{x}\right) + 5\left(1 - \frac{1}{x}\right)^2 + 7\left(1 - \frac{1}{x}\right)^3 + \dots \text{ is}$$

$$2x^2 - x$$

Statement-2: If $0 < y < 1$, the sum of the

$$\text{series } 1 + 3y + 5y^2 + 7y^3 + \dots \text{ is } \frac{1+y}{(1-y)^2}$$

- 1) 1 2) 2 3) 3 4) 4

6. **Let** a, b, c **be positive real numbers in H.P.**

$$\text{Statement-1: } \frac{a+b}{2a-b} + \frac{c+b}{2c-b} \geq 4$$

$$\text{Statement-2: } \frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$$

- 1) 1 2) 2 3) 3 4) 4

7. Suppose four distinct positive numbers a_1, a_2, a_3, a_4 are in G.P. Let

$$b_1 = a_1, b_2 = b_1 + a_2, b_3 = b_2 + a_3 \text{ and } b_4 = b_3 + a_4.$$

Statement-1: The numbers b_1, b_2, b_3, b_4 are neither in A.P. nor in G.P. and

Statement-2: The numbers b_1, b_2, b_3, b_4 are in H.P.

- 1) 1 2) 2 3) 3 4) 4

8. **Statement-1:** If a, b, c are distinct real numbers in H.P, then $a^n + c^n > 2b^n, \forall n \in N$

Statement-2: A.M > G.M > H.M

- 1) 1 2) 2 3) 3 4) 4

- 9 **Statement-1:**

$$\frac{1^2}{1.3} + \frac{2^2}{3.5} + \frac{3^2}{5.7} + \dots + \frac{n^2}{(2n-1)(2n+1)} = \frac{n(n+1)}{2(2n+1)}$$

Statement-2:

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{1}{2n+1}$$

- 1) 1 2) 2 3) 3 4) 4

10. Let $n \in N$ and k be an integer ≥ 0 such that $S_k(n) = 1^k + 2^k + 3^k + \dots + n^k$

Statement-I:

$$S_4(n) = \frac{n}{30}(n+1)(2n+1)(3n^2+3n+1)$$

Statement-II:

$${}^{k+1}C_1 S_k(n) + {}^{k+1}C_2 S_{k-1}(n) + \dots + {}^{k+1}C_k S_1(n) + {}^{k+1}C_{k+1} S_0(n) = (n+1)^{k+1} - 1$$

- 1) 1 2) 2 3) 3 4) 4

11. Let S_n denote the sum of n terms of the series

$$1^2 + 3 \times 2^2 + 3^2 + 3 \times 4^2 + 5^2 + 3 \times 6^2 + 7^2 + \dots$$

Statement-1: If n is odd, then

$$S_n = \frac{n(n+1)(4n-1)}{6}$$

Statement-2: If n is even, then

$$S_n = \frac{n(n+1)(4n+5)}{6}$$

- 1) 1 2) 2 3) 3 4) 4

12. **Statement 1:** The sum of series $1 + (1+2+4) + (4+6+9) + (9+12+16) + \dots + (361+380+400)$ is 8000.

Statement 2: $\sum_{k=1}^n (k^3 - (k-1)^3) = n^3$ for any

natural number n. [AIEEE 2012]

- 1) 1 2) 2 3) 3 4) 4

LEVEL-IV - KEY

- 1) 1 2) 1 3) 1 4) 1 5) 1 6) 2
7) 3 8) 1 9) 3 10) 4 11) 2 12) 1

LEVEL-IV - HINTS

1. Statement -2 is a property of an A.P. So, it is true. Now,

$$a_1 + a_4 + a_7 + \dots + a_{16} = 147$$

$$\Rightarrow (a_1 + a_{16}) + (a_4 + a_{13}) + (a_7 + a_{10}) = 147$$

$$\Rightarrow 3(a_1 + a_{16}) = 147 \quad [\text{Using statement-2}]$$

$$\Rightarrow a_1 + a_{16} = 49$$

$$\therefore a_1 + a_6 + a_{11} + a_{16}$$

$$= (a_1 + a_{16}) + (a_6 + a_{11}) = 2(a_1 + a_{16}) = 2 \times 49 = 98$$

2. Clearly, statement-2 is true.

Let $S_n = a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$. Then,

$$S_n = a_n + a_{n-1} + a_{n-2} + \dots + a_2 + a_1$$

$$2S_n = (a_1 + a_n) + (a_2 + a_{n-1}) + (a_3 + a_{n-2}) + \dots + (a_n + a_1)$$

$$\Rightarrow 2S_n = n(a_1 + a_n) \quad [\text{Using statement-2}]$$

$$\Rightarrow S_n = \frac{n}{2}(a_1 + a_n)$$

So, statement-2 is correct explanation for statement-1.

3. Suppose a_p, a_q and a_r are $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms of an A.P. whose common difference is d . Then, $a_p - a_r = (p-r)d$ and

$$a_p - a_q = (p-q)d$$

$$\therefore \frac{a_p - a_q}{a_p - a_r} = \frac{p-q}{p-r}, \text{ which is a rational number.}$$

So, statement-2 is true.

In order to check the truth of statement-1, let us assume that there is an A.P. having $\sqrt{3}, \sqrt{5}$ and $\sqrt{7}$ as its p^{th}, q^{th} and r^{th} terms. Then, $\frac{\sqrt{3}-\sqrt{5}}{\sqrt{3}-\sqrt{7}}$ should be a rational number.

This is a contradiction.

Hence, $\sqrt{3}, \sqrt{5}$ and $\sqrt{7}$ cannot be three terms of an A.P. So, statement-1 is true and statement-2 is a correct explanation for statement-1.

4. As $2^{1+x} + 2^{1-x}, \lambda$ and $3^x + 3^{-x}$ are three equidistant terms of an AP.

$\therefore 2^{1+x} + 2^{1-x}, \lambda$ and $3^x + 3^{-x}$ are in A.P.

$$\Rightarrow 2\lambda = 2^{1+x} + 2^{1-x} + 3^x + 3^{-x}$$

$$\Rightarrow 2\lambda = 2(2^x + 2^{-x}) + (3^x + 3^{-x})$$

Now, A.M > G.M

$$\Rightarrow 2^x + 2^{-x} \geq 2 \text{ and } 3^x + 3^{-x} \geq 2$$

$$\Rightarrow 2(2^x + 2^{-x}) + (3^x + 3^{-x}) \geq 4 + 2 \Rightarrow 2\lambda \geq 6 \Rightarrow \lambda \geq 3$$

5. Let $S = 1 + 3y + 5y^2 + 7y^3 + \dots$ (i)

$$\text{Then, } yS = y + 3y^2 + 5y^3 + \dots \quad \text{(ii)}$$

Subtracting (ii) from (i), we get

$$S(1-y) = 1 + 2y + 2y^2 + 2y^3 + \dots$$

$$\Rightarrow S(1-y) = 1 + \frac{2y}{(1-y)} \Rightarrow S = \frac{1+y}{(1-y)^2}$$

So, statement-2 is true.

Putting $y = 1 - \frac{1}{x}$ in the above sum, we get

$$S = 2x^2 - x$$

So, statement-1 is also true and statement-2 is a correct explanation for statement-1.

6. Since a, b, c are in H.P. $\therefore b = \frac{2ac}{a+c}$. Now,

$$\frac{a+b}{2a-b} + \frac{c+b}{2c-b} = \frac{a + \frac{2ac}{a+c}}{2a - \frac{2ac}{a+c}} + \frac{c + \frac{2ac}{a+c}}{2c - \frac{2ac}{a+c}}$$

$$\frac{a+b}{2a-b} + \frac{c+b}{2c-b} = 1 + \frac{3}{2} \left(\frac{a}{c} + \frac{c}{a} \right)$$

$$\frac{a+b}{2a-b} + \frac{c+b}{2c-b} \geq 1 + 3 = 4$$

$$\left[A.M \geq G.M, \frac{a}{c} + \frac{c}{a} \geq 2 \right]$$

So, statement-2 is true,

But, statement-2 is not a correct explanation for statement-1.

7. Let r be the common ratio of the G.P. We have,

$$b_1 = a_1, b_2 = a_1 + a_2 = a_1(1+r), b_3 = b_2 + a_3 = a_1(1+r+r^2),$$

$$b_4 = b_3 + a_4 = a_1(1+r+r^2+r^3)$$

Clearly, also b_1, b_2, b_3, b_4 are neither in A.P nor in H.P. so, statement-1 is true.

8. If a, b, c are in H.P., then b is the H.M of a and c . But, the G.M of a and c is $\sqrt{ac}$.

$$\therefore G.M > H.M \Rightarrow \sqrt{ac} > b \text{ --- (i)}$$

Using A.M > H.M, we have

$$\frac{a^n + c^n}{2} > \sqrt{a^n c^n}$$

$$\Rightarrow \frac{a^n + c^n}{2} > (ac)^{n/2}$$

$$\Rightarrow \frac{a^n + c^n}{2} > b^n \quad \text{[Using (i)]}$$

$$\Rightarrow a^n + c^n > 2b^n$$

Hence, statement-1 and 2 are true and statement-2 is a correct explanation for statement-1

9. Verification with $n=2$
 10. Verification with $n=2$
 11. Both statements are true but 2nd statement is not a correct exp. for statement 1
 12. $T_n = (n-1)^2 + (n-1)n + n^2$

$$= \frac{((n-1)^3 - n^3)}{(n-1) - n} = n^3 - (n-1)^3$$

Find $T_1, T_2, \dots, T_{20}$ and $S_{20} = 20^3 - 0^3 = 8000$

